Name

Mathematics Stage 5 Paths

Further Non-Linear Relationships

Book 1 Transformations of exponentials, hyperbolas, and circles.
Solving equations graphically.

Version: 261007

Feedback:
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Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-NLI-P-01 interprets and compares non-linear relationships and their transformations, both algebraically and graphically

Graph exponentials and describe their features and transformations

- Use graphing applications to graph exponential relationships of the form $y = k(a)^x + c$ and $y = k(a)^{-x} + c$ for integer values of k, a and c (where $a > 0$ and $a \neq 1$), and compare and describe any relevant features

Graph hyperbolas and describe their features and transformations

- Use graphing applications to graph, compare and describe hyperbolic relationships of the form $y = \frac{k}{x}$ for integer values of k
- Use graphing applications to graph and describe a variety of hyperbolas, including where the equation is given in the form $y = \frac{k}{x} + c$ or $y = \frac{k}{x - b}$ for integer values of k, b and c

Graph circles and describe their features and transformations

- Derive the equation of a circle $x^2 + y^2 = r^2$ with centre $(0, 0)$ and radius r using the distance formula
- Identify and describe equations that represent circles with centre at the origin and radius of the circle r
- Graph circles of the form $x^2 + y^2 = r^2$, where r is the radius of the circle using graphing applications
- Establish the equation of the circle with centre (a, b) and radius r, and graph equations of the form $(x - a)^2 + (y - b)^2 = r^2$
- Find the centre and radius of a circle with the equation in the form $x^2 + y^2 + ax + by + c = 0$ by completing the square

Distinguish between different types of graphs by examining their algebraic and graphical representations and solve problems

- Identify and describe features of different types of graphs based on their equations
- Identify a possible equation from a graph and verify using graphing applications
- Find points where a line intersects with a parabola, hyperbola or circle, both graphically and algebraically

Graph and compare polynomial curves and describe their features and transformations

- Use graphing applications to graph and compare features of cubic equations of the form $y = ax^3 + c$, where a and c are integers
- Use graphing applications to graph a variety of equations of the form $y = kx^n$, where n is an integer and $n \geq 2$, and describe the effect on the shape of the curve where n is an odd or an even number
- Use graphing applications to graph curves of the form $y = kx^n + c$ and $y = k(x - b)^n$ where n is an integer and $n \geq 2$, and describe the transformations from $y = kx^n$

Exponential Curve Transformations

ⓘ Exponential Curve Transformations

$a > 1$ exponential growth
$0 < a < 1$ exponential decay
Common ratio

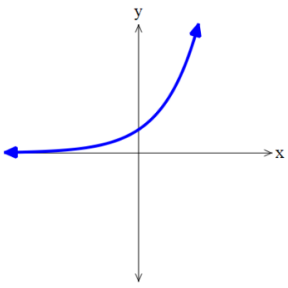
$$y = k \cdot a^x$$

Factor; affects y-intercept and reflects across x axis if negative

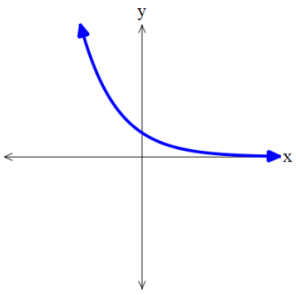
$a > 1$ exponential decay
$0 < a < 1$ exponential growth
Common ratio

$$y = k \cdot a^{-x}$$

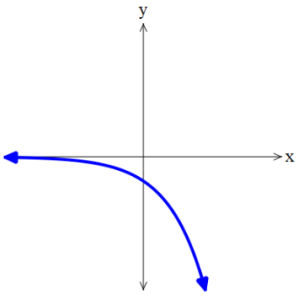
$y = 2^x$



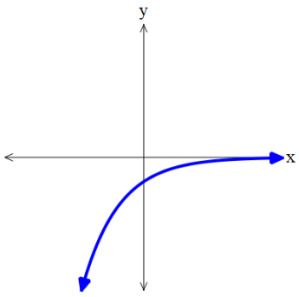
$y = 2^{-x}$ or $y = \left(\frac{1}{2}\right)^x$



$y = -2^x$



$y = -2^{-x}$ or $y = -\left(\frac{1}{2}\right)^x$



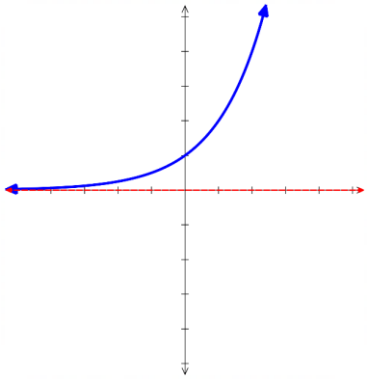
ⓘ Exponential Curve Transformations

Consider the equation of an exponential curve $y = k(a)^x + c$, where $a > 0$

The parameters a, c, k are numbers that affect the graph in predictable ways.

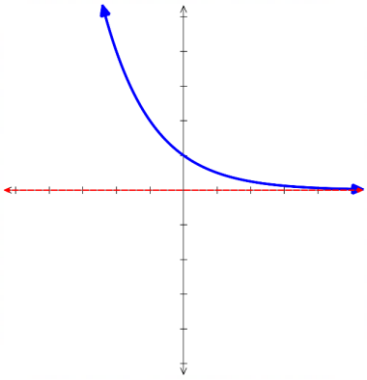
$a > 1$, we call this exponential ……

$y = 2^x$



$0 < a < 1$, we call this exponential ……

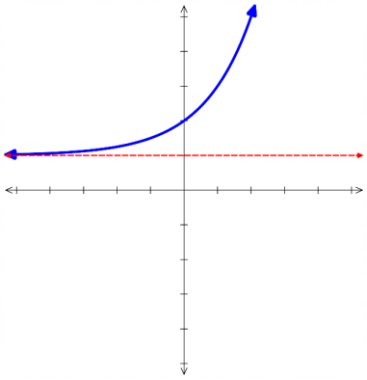
$y = \left(\frac{1}{2}\right)^x$



$c > 0$, the curve is shifted ……

The value of c is the ……

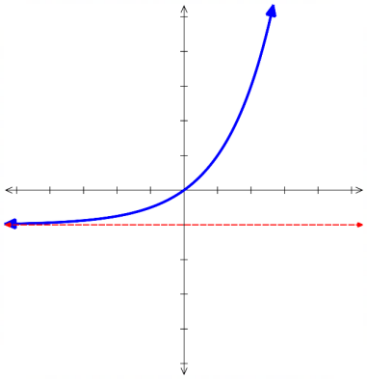
$y = 2^x + 1$



$c < 0$, the curve is shifted ……

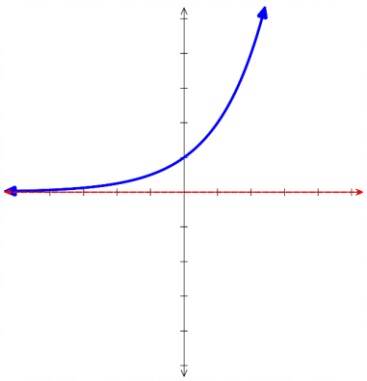
The value of c is the ……

$y = 2^x - 1$



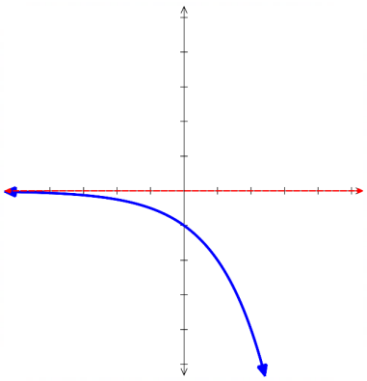
$k > 0$

$y = 1(2^x)$



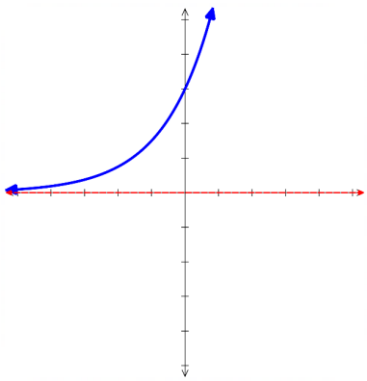
$k < 0$, reflect the graph across x-axis

$y = -1\ (2^x)$



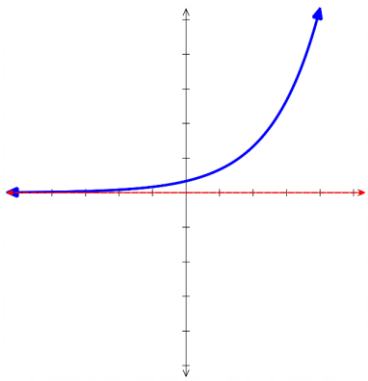
$|k| > 1$, the curve changes more quickly.

$y = 3(2^x)$



$0 < |k| < 1$ the curve changes more slowly.

$y = \frac{1}{3}(2^x)$



ⓘ Exponential Curve Transformations

$a > 1$	exponential growth
$0 < a < 1$	exponential decay
	Common ratio

$$y = k \cdot a^x + c$$

Factor; affects y-intercept and reflects across x axis if negative

vertical translation $y = c$ is asymptote

$a > 1$	exponential decay
$0 < a < 1$	exponential growth
	Common ratio

$$y = k \cdot a^{-x} + c$$

To sketch exponential curves:

1. Find y-intercept by letting $x = 0$.
2. Find asymptote: $y = c$.
3. Plot at least one other point, such as at $x = 1$.

⚠ You do not need to find x-intercepts until you learn logarithms

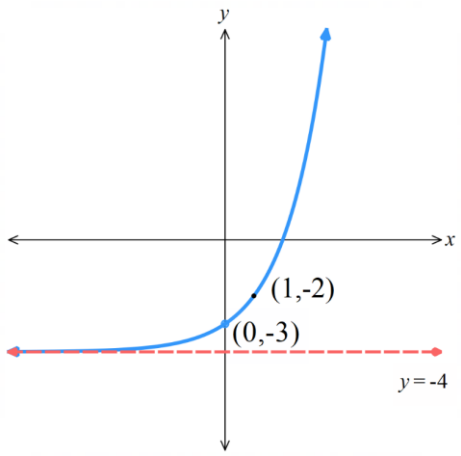
Example Sketch transformations of exponential curves

$y = 2^x - 4$

1. y-int: Let $x = 0$, $y = 2^0 - 4 = -3$

2. Asymptote: $y = -4$

3. When $x = 1$: $y = 2^1 - 4$, $y = -2$

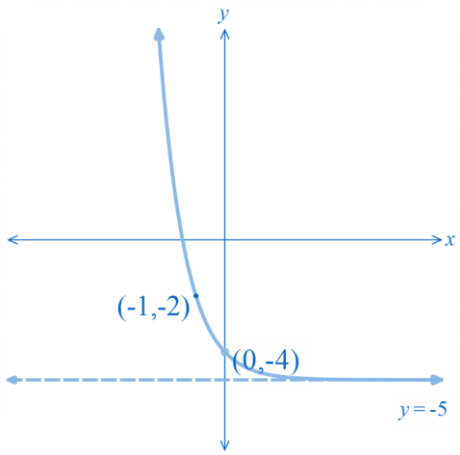


$y = 3^{-x} - 5$

1. y-int:

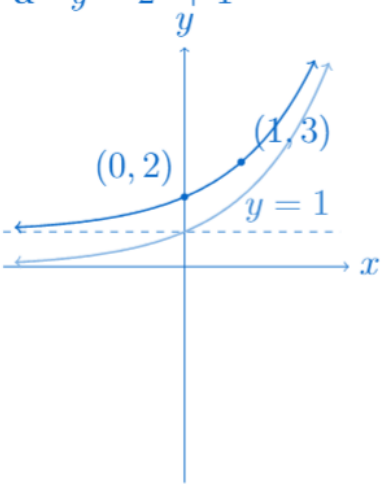
2. Asymptote:

3. Another point:

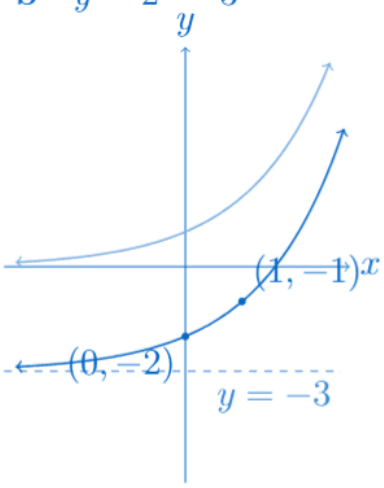


1 The graph of $y = 2^x$ is given. Sketch these graphs on the same axis to show the transformation. Clearly label the y-intercept, asymptote, and the point where $x = 1$.

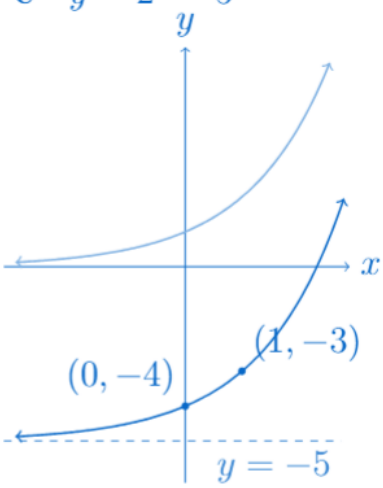
a $y = 2^x + 1$



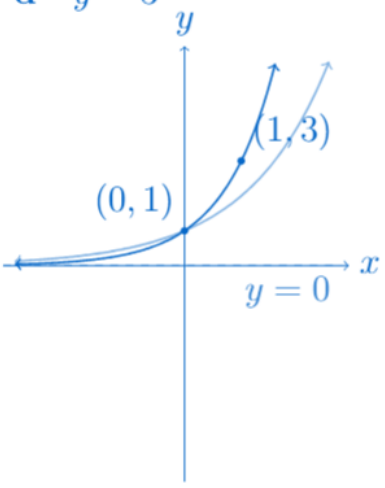
b $y = 2^x - 3$



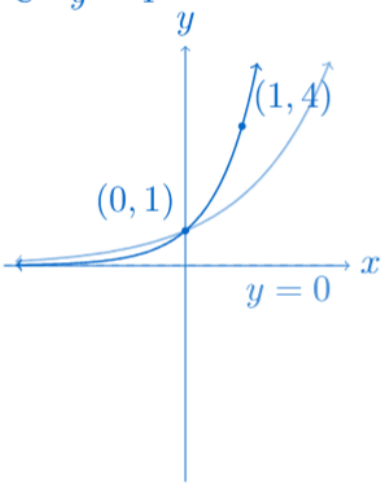
c $y = 2^x - 5$



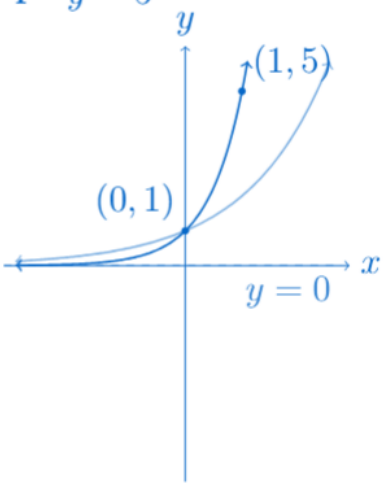
d $y = 3^x$



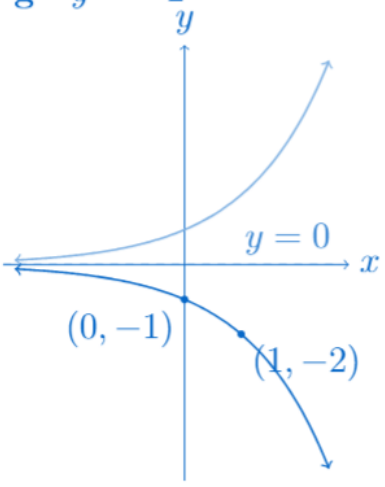
e $y = 4^x$



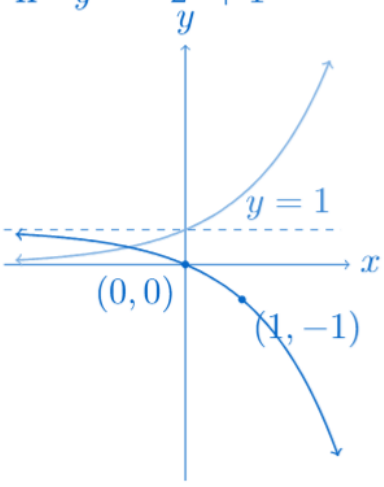
f $y = 5^x$



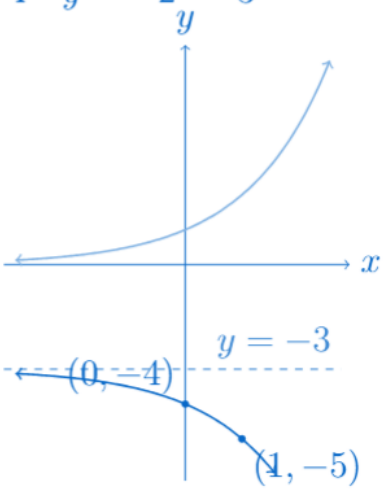
g $y = -2^x$



h $y = -2^x + 1$



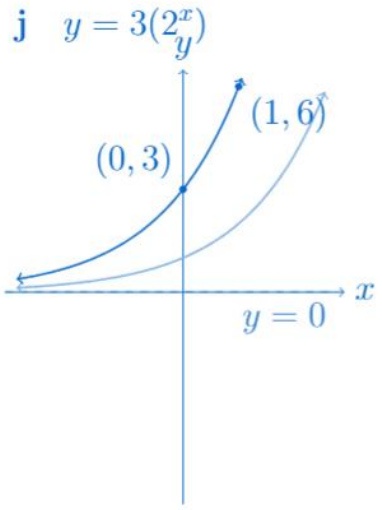
i $y = -2^x - 3$

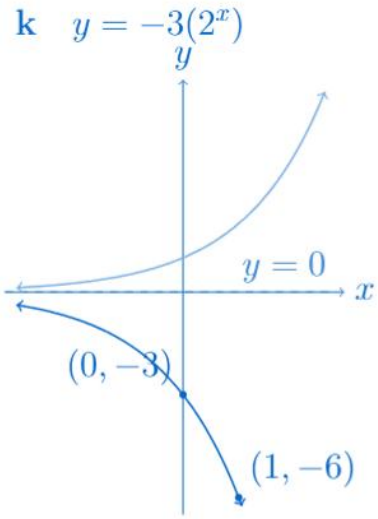


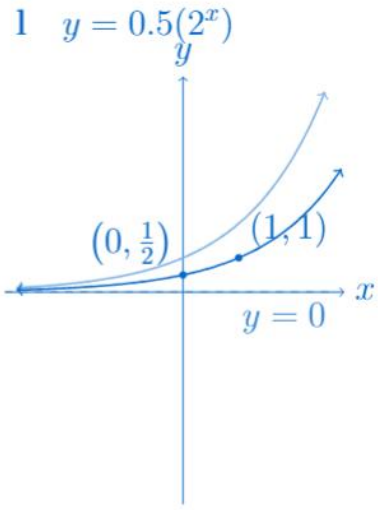
j $y = 3(2^x)$

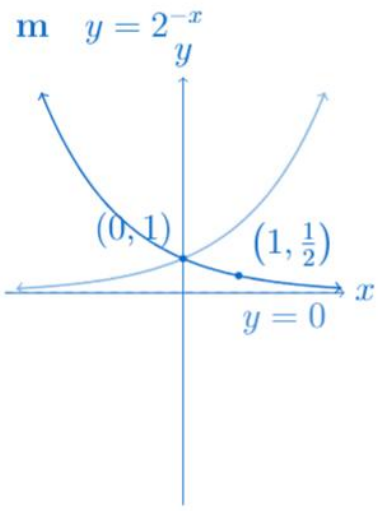
k $y = -3(2^x)$

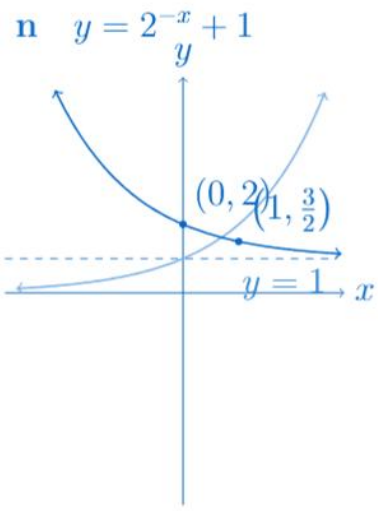
l $y = 0.5(2^x)$

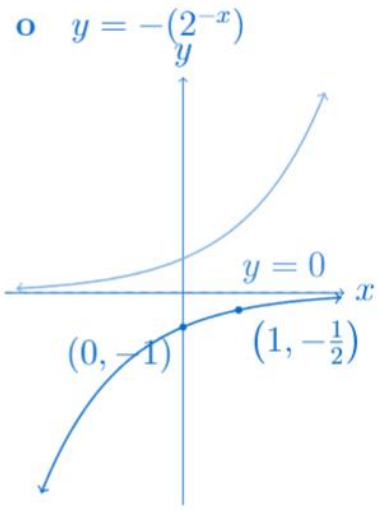


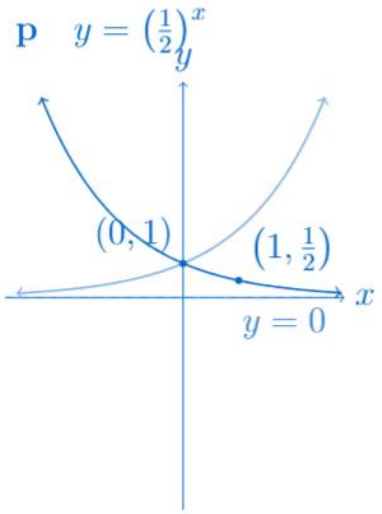


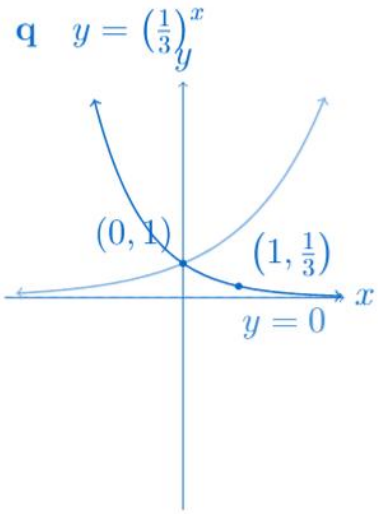


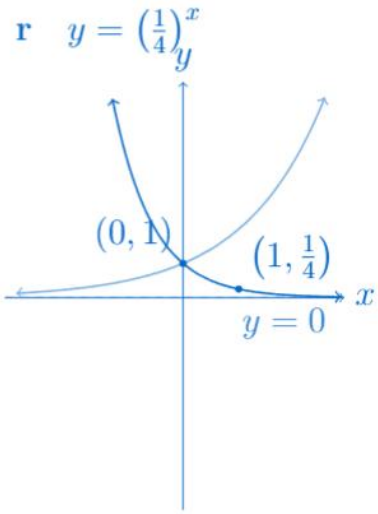






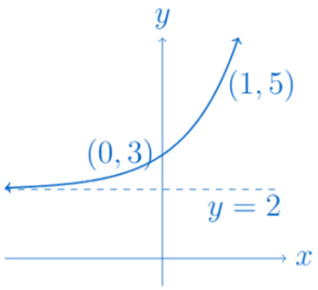




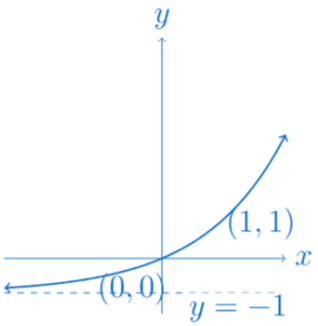


2 Sketch these exponential equations.
Clearly label the y-intercept, asymptote, and the point where $x = 1$.

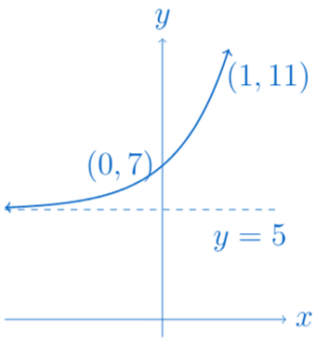
a $y = 3^x + 2$



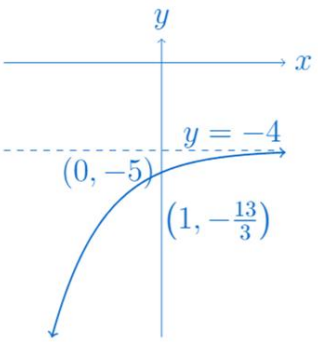
b $y = 2^x - 1$



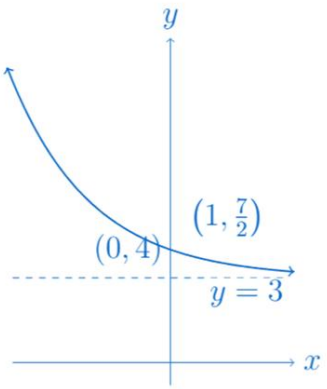
c $y = 2(3^x) + 5$



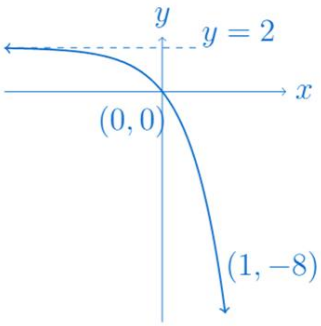
d $y = -3^{-x} - 4$



e $y = 2^{-x} + 3$



f $y = -2(5^x) + 2$

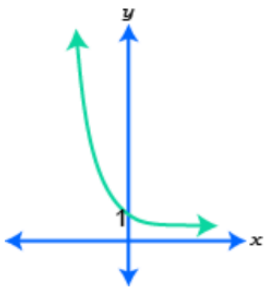


3 Match the equation to the graph.

$y = 6^x$ $y = -6^x$ $y = -6^x - 2$ $y = -6^x + 2$ $y = 6^{-x}$

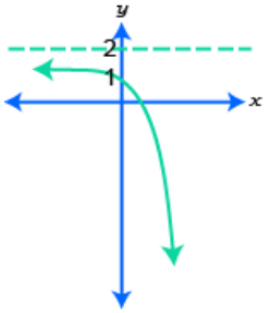
$y = 6^{-x} - 2$ $y = 6^{-x} + 2$ $y = -6^{-x}$ $y = -6^{-x} - 2$ $y = 6^{-x} + 2$

a



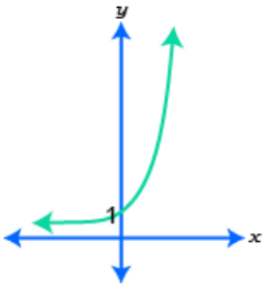
$y = 6^{-x}$

b



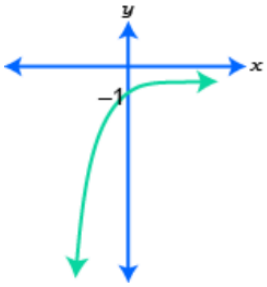
$y = -6^x + 2$

c



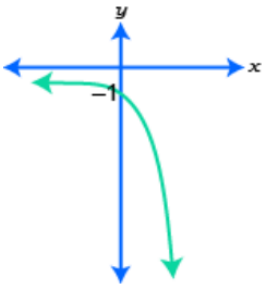
$y = 6^x$

d



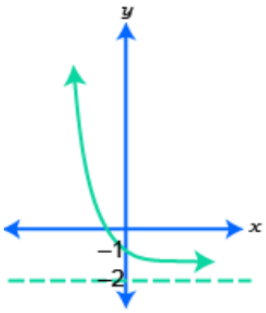
$y = -6^{-x}$

e



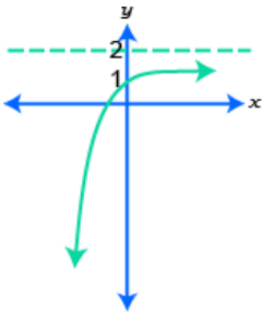
$y = -6^x$

f



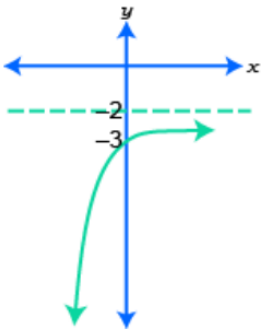
$y = 6^{-x} - 2$

g



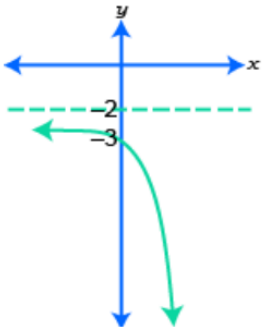
$y = -6^{-x} + 2$

h



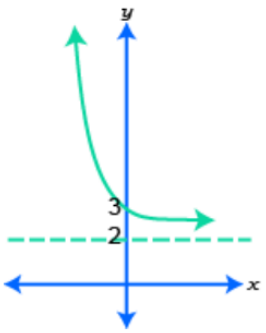
$y = -6^{-x} - 2$

i



$y = -6^x - 2$

j



$y = 6^{-x} + 2$

Hyperbolas

ⓘ Reciprocal Relationships

Reciprocal relationships are in the form $y = \frac{k}{x}$.

Their graph will not be a straight line; so the relationship is **non-linear.**

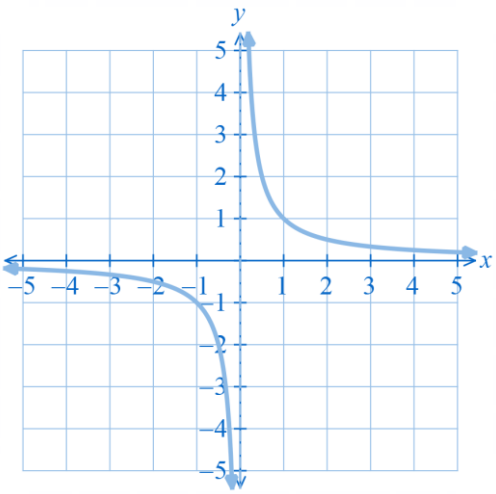
We can graph reciprocal relationships using these steps:

1. Construct a table of values using the equation.
2. Graph the table by using the columns as coordinates.
3. Join the points to form a smooth curve.

ⓘ **Investigation** Graph reciprocal relationships using a table of values

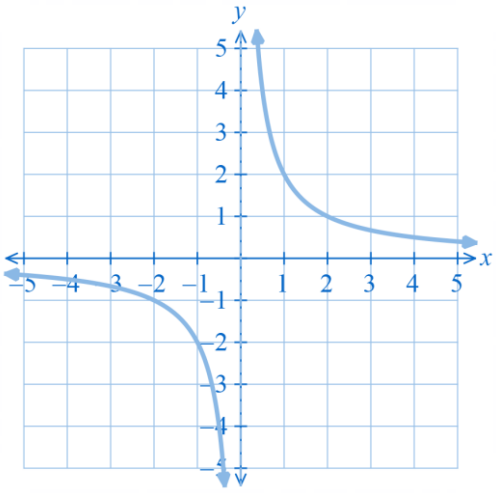
Draw the graph of $y = \frac{1}{x}$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							



Draw the graph of $y = \frac{2}{x}$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							

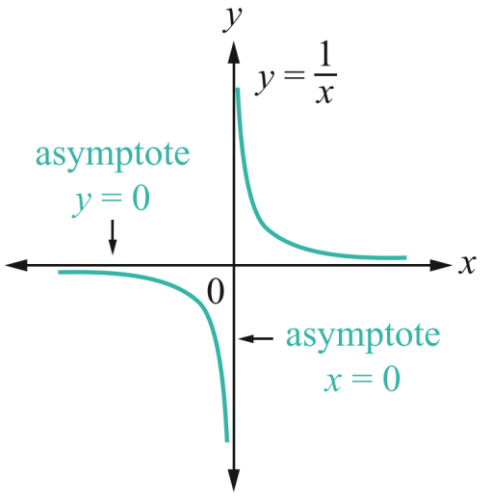


ⓘ Properties of Hyperbolas

A **reciprocal** relationship $y = \frac{k}{x}$ has a graph in the shape of a **hyperbola**.

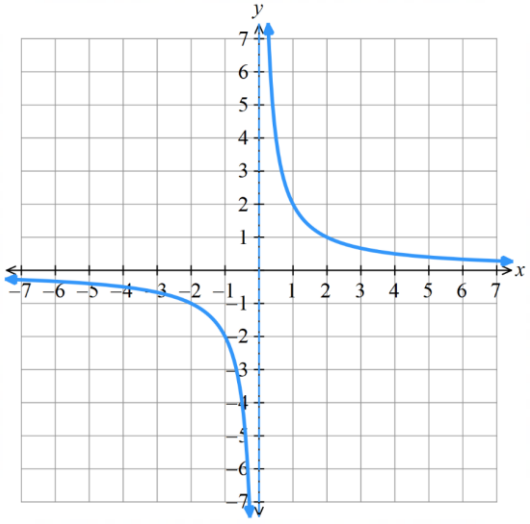
Hyperbolas have:

- Two **asymptotes**: lines where the graph approaches but never reaches.
- No x or y-intercepts.
- Two symmetrical branches.



ⓘ **Identify** Features of hyperbolas

a $y = \frac{2}{x}$



Identify the asymptotes.

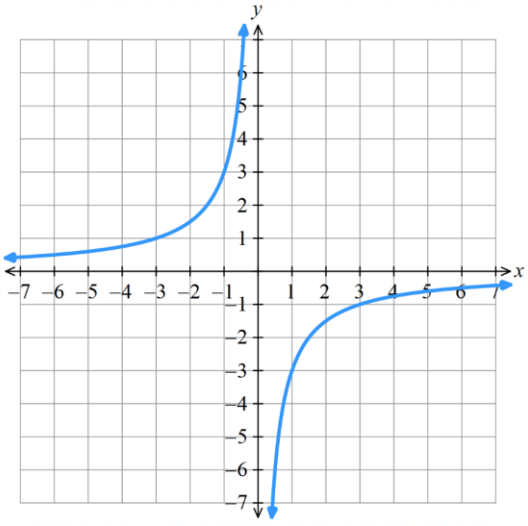
Behaviour as $x \to \infty$:

Behaviour as $x \to -\infty$:

Behaviour as $x \to 0^+$:

Behaviour as $x \to 0^-$:

b $y = -\frac{2}{x}$



Identify the asymptotes.

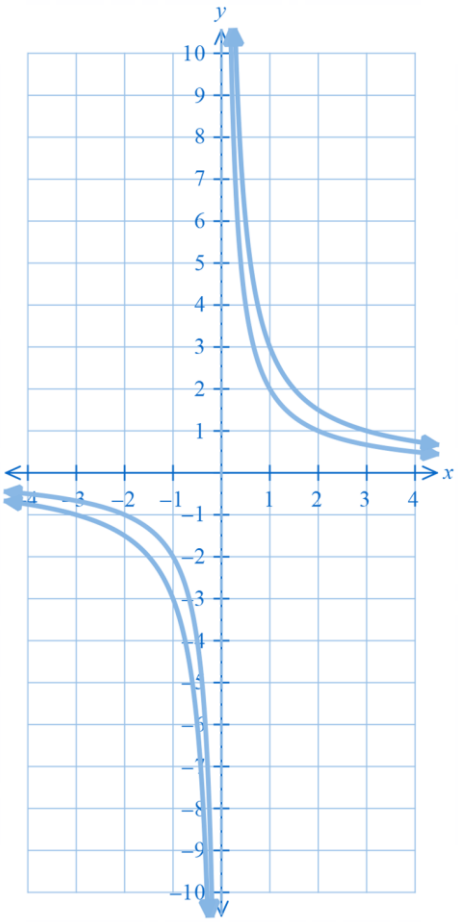
Behaviour as $x \to \infty$:

Behaviour as $x \to -\infty$:

Behaviour as $x \to 0^+$:

Behaviour as $x \to 0^-$:

1 Plot $y = \frac{2}{x}$ and $y = \frac{3}{x}$ on the same axis.



$y = \frac{2}{x}$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							

Behaviour as $x \to \infty$:

Behaviour as $x \to -\infty$:

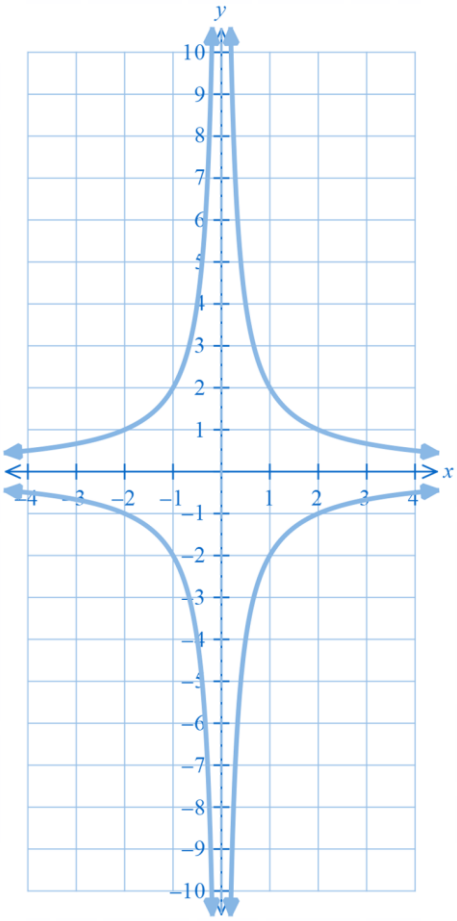
$y = \frac{3}{x}$

x	-2	-1	$-\frac{1}{3}$	0	$\frac{1}{3}$	1	2
y							

Behaviour as $x \to \infty$:

Behaviour as $x \to -\infty$:

2 Plot these graphs on the same axis.



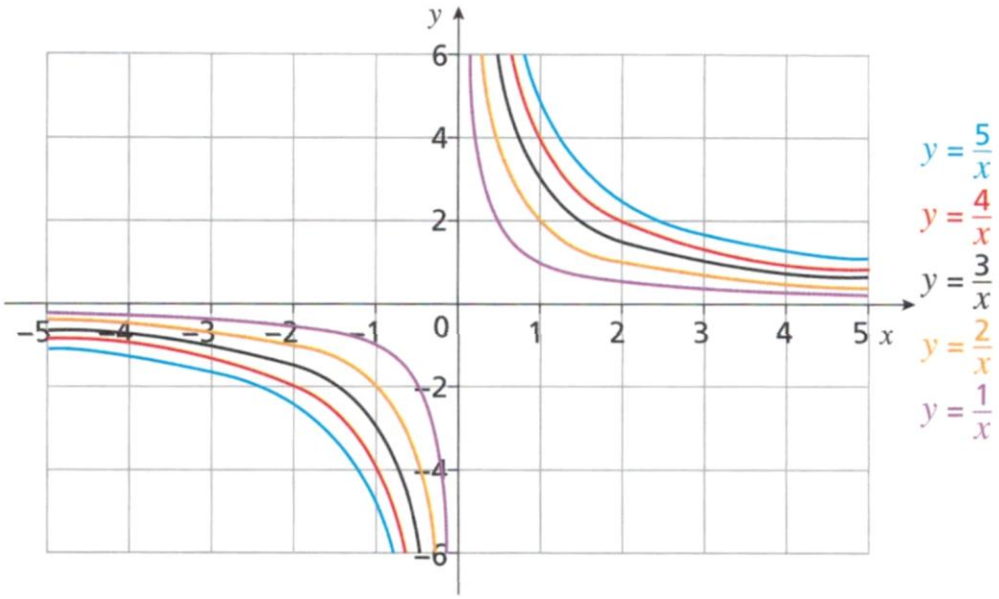
x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
$y = \frac{2}{x}$							
$y = -\frac{2}{x}$							

Describe their differences in terms of:

Behaviour as $x \to \infty$:

Behaviour as $x \to -\infty$:

3 Here are the graphs of $y = \frac{1}{x}$, $y = \frac{2}{x}$, $y = \frac{3}{x}$, $y = \frac{4}{x}$ and $y = \frac{5}{x}$

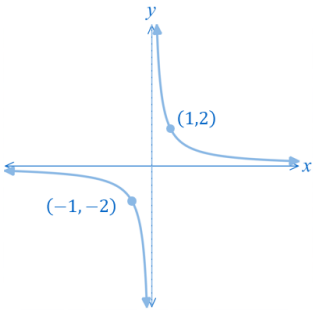


What's the same and what's different about the graphs?

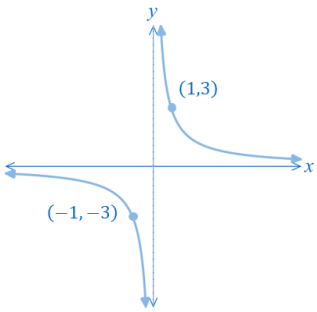
Same:	Different:
Asymptotes at $x = 0$ and $y = 0$, in quadrants 1 and 3.	For $y = \frac{k}{x}$, the larger the value of k, the further away from the axes.

4 Sketch these hyperbolas, labelling the point where $x = 1$ and $x = -1$.

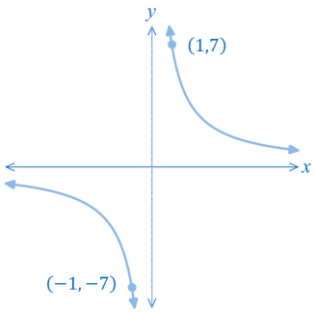
a $y = \frac{2}{x}$



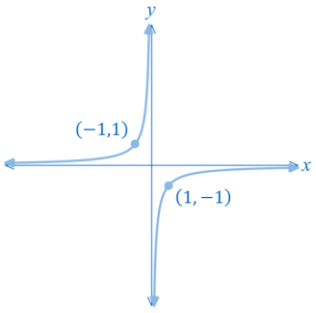
b $y = \frac{3}{x}$



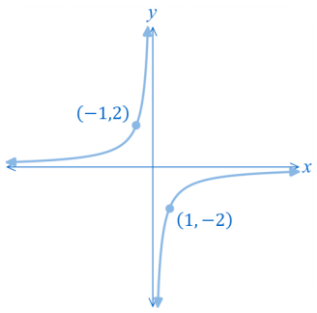
c $y = \frac{7}{x}$



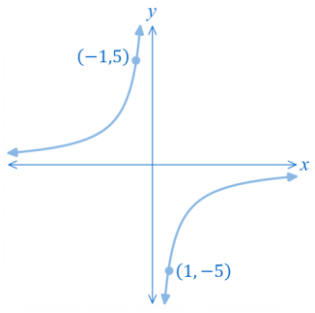
d $y = -\frac{1}{x}$



e $y = -\frac{2}{x}$



f $y = -\frac{5}{x}$



Hyperbola Transformations

Investigation Equation and the graph of a hyperbola

https://www.desmos.com/calculator

a Sketch these graphs

$y = \frac{1}{x}$

$y = \frac{2}{x}$

$y = \frac{3}{x}$

$y = \frac{4}{x}$

Increasing the value of k s…… or *dilates* the hyperbola away from the axes.

b Sketch these graphs.

$y = \frac{-1}{x}$

$y = \frac{-2}{x}$

$y = \frac{-3}{x}$

$y = \frac{-4}{x}$

A negative value of k means the hyperbola is in the …… and …… quadrants.

c Sketch these graphs

$y = \frac{1}{x} + 1$

$y = \frac{1}{x} + 2$

$y = \frac{1}{x} + 3$

$y = \frac{1}{x} - 3$

$y = \frac{1}{x} - 4$

The constant term shifts the hyperbola v……

d Sketch these graphs

$y = \frac{1}{x+1}$

$y = \frac{1}{x+2}$

$y = \frac{1}{x+3}$

$y = \frac{1}{x-3}$

$y = \frac{1}{x-4}$

The constant term with the x shifts the hyperbola h……

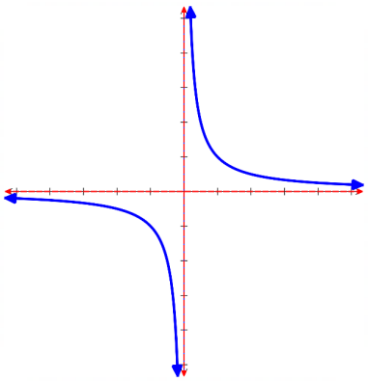
ⓘ Hyperbola Transformations

Consider the equation of a hyperbola $y = \frac{k}{x-b} + c$, where k is an integer.

The parameters k, b, c are numbers that affect the graph in predictable ways.

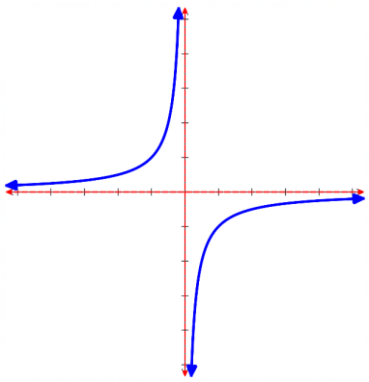
k is positive, the hyperbola is top …… and bottom …… relative to the asymptotes.

$y = \frac{1}{x}$



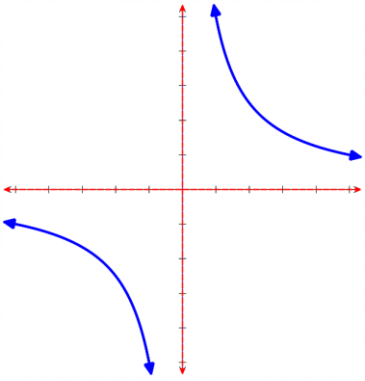
k is negative, the hyperbola is …… and …… relative to the asymptotes.

$y = -\frac{1}{x}$

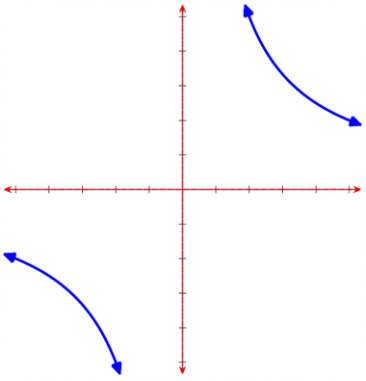


As $|k|$ increases, the hyperbola gets *dilated*; it stretches away from the axes.

$y = \frac{5}{x}$

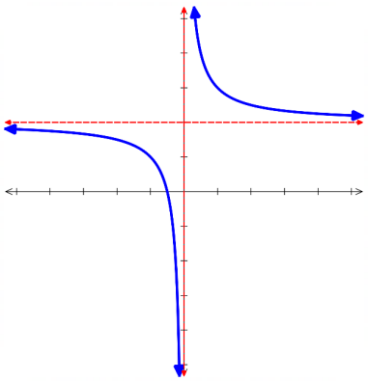


$y = \frac{10}{x}$

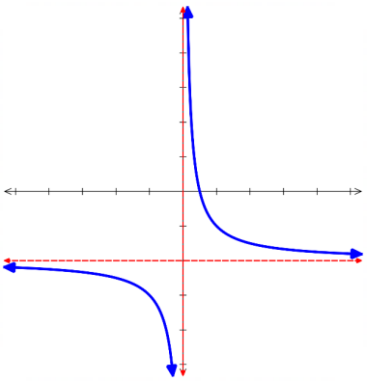


c shifts the curve …… The value of c is the …… asymptote.

$y = \frac{1}{x} + 2$

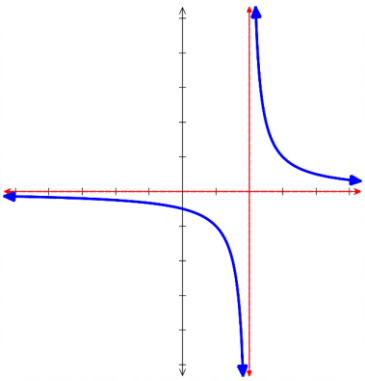


$y = \frac{1}{x} - 2$

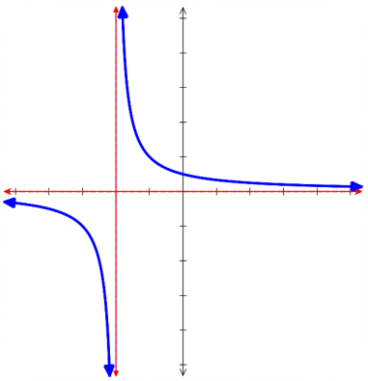


b shifts the curve …… The value of b is the …… asymptote.

$y = \frac{1}{x-2}$



$y = \frac{1}{x+2}$



Sketching Hyperbolas

Dilates, or stretches, hyperbola

$$y = \frac{k}{x-b} + c$$

Horizontal translation: $x = b$ is a vertical asymptote

Vertical translation: $y = c$ is a horizontal asymptote

To sketch hyperbolas:

1. Find y-intercept, if any, by letting $x = 0$.
2. Find x-intercept, if any, by letting $y = 0$.
3. Find vertical asymptote at $x = b$ and horizontal asymptote at $y = c$,

Example Sketch hyperbolas, showing intercepts and asymptotes

$$y = \frac{1}{x+1} + 2$$

1. y-int: Let $x = 0$,

$$y = \frac{1}{1} + 2 = 3$$

2. x-int: Let $y = 0$

$$0 = \frac{1}{x+1} + 2$$

$$-2 = \frac{1}{x+1}$$

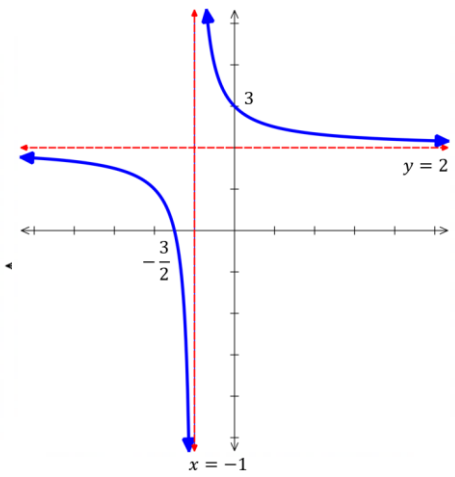
$$-2x - 2 = 1$$

$$-2x = 3$$

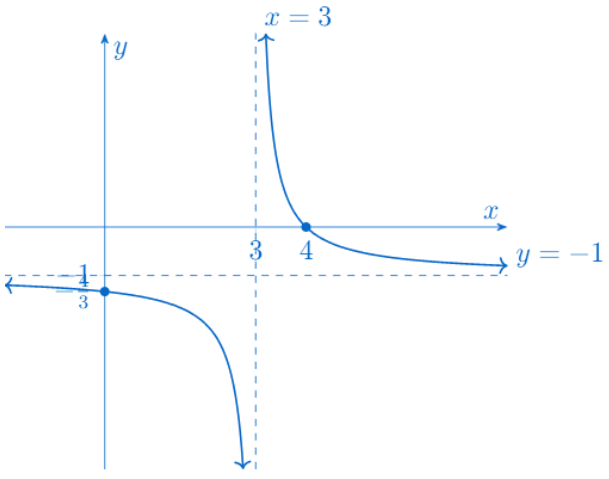
$$x = -\frac{3}{2}$$

3. Asymptotes: $x = -1$

$y = 2$



$$y = \frac{1}{x-3} - 1$$



1 Describe these translations.

a Compared with $y = \frac{1}{x}$, $y = \frac{1}{x-3}$ is translated right by 3

b Compared with $y = \frac{1}{x}$, $y = \frac{1}{x+2}$ is translated .....

Left 2

c Compared with $y = \frac{1}{x}$, $y = \frac{1}{x} + 3$ is translated .....

Up 3

d Compared with $y = \frac{1}{x}$, $y = \frac{1}{x} - 6$ is translated .....

Down 6

e Compared with $y = \frac{1}{x}$, $y = \frac{1}{x+3} - 2$ is translated .....

Left 3, down 2

f Compared with $y = \frac{1}{x}$, $y = \frac{1}{x+5} + 7$ is translated .....

Left 5, up 7

g Compared with $y = \frac{1}{x}$, $y = \frac{1}{x-5} + 3$ is translated .....

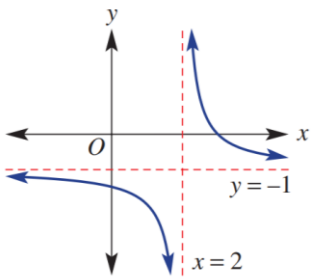
Right 5, up 3

h Compared with $y = \frac{1}{x}$, $y = \frac{1}{x+1} - 9$ is translated .....

Left 1, down 9

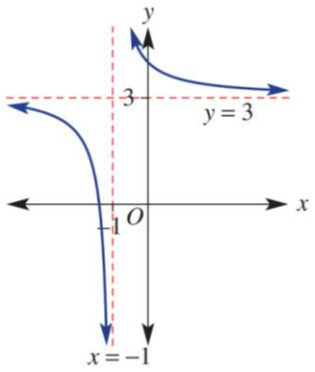
2 Determine the equation of these hyperbolas in the form $y = \pm\frac{1}{x-b} + c$

a



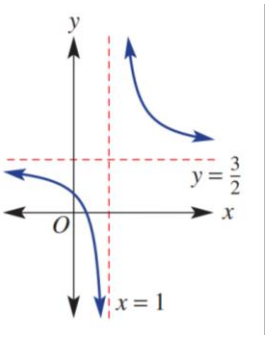
$y = \frac{1}{x-2} - 1$

b



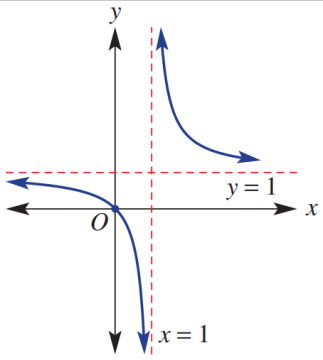
$y = \frac{1}{x+1} + 3$

c



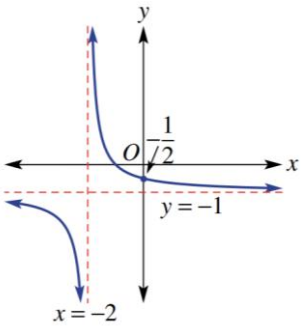
$y = \frac{1}{x-1} + \frac{3}{2}$

d



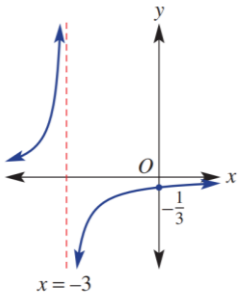
$y = \frac{1}{x-1} + 1$

e



$y = \frac{1}{x+2} - 1$

f

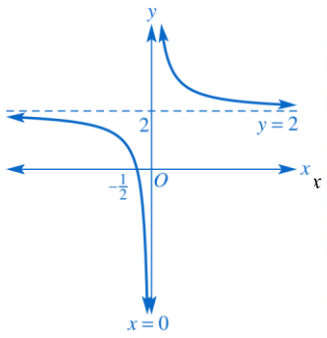


$y = -\frac{1}{x+3}$

3 Sketch these graphs on the same axis to show the transformation.
Clearly label the asymptotes and intercepts.

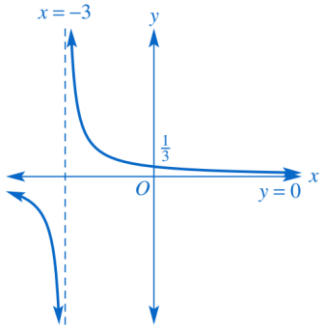
a

$$y = \frac{1}{x} + 2$$



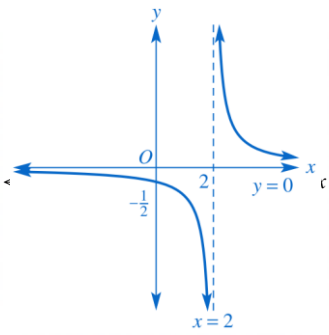
b

$$y = \frac{1}{x+3}$$



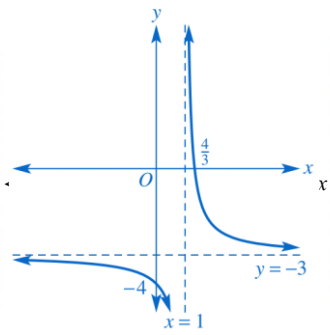
c

$$y = \frac{1}{x-2}$$



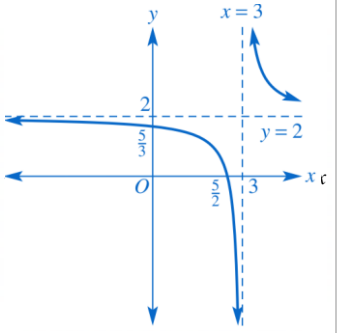
d

$$y = \frac{1}{x-1} - 3$$



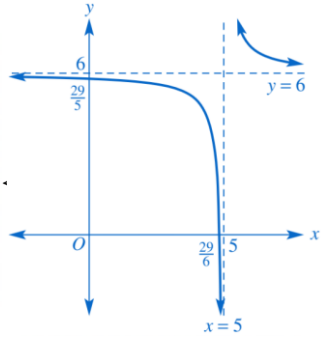
e

$$y = \frac{1}{x-3} + 2$$



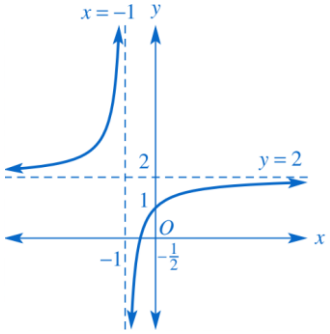
f

$$y = \frac{1}{x-5} + 6$$



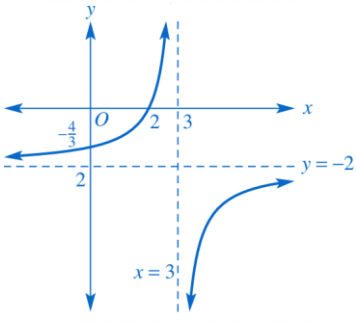
g

$$y = \frac{-1}{x+1} + 2$$



h

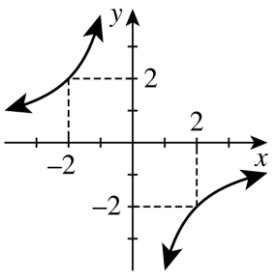
$$y = \frac{-2}{x-3} - 2$$



4 These hyperbolas have equations in the form $y = \frac{k}{x}$.

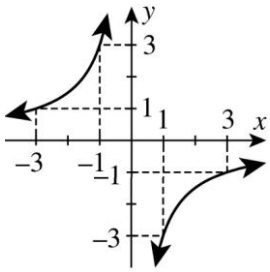
Determine the value of k by substituting in a coordinate and hence find the equation of the hyperbola.

a



$y = -\frac{4}{x}$

b

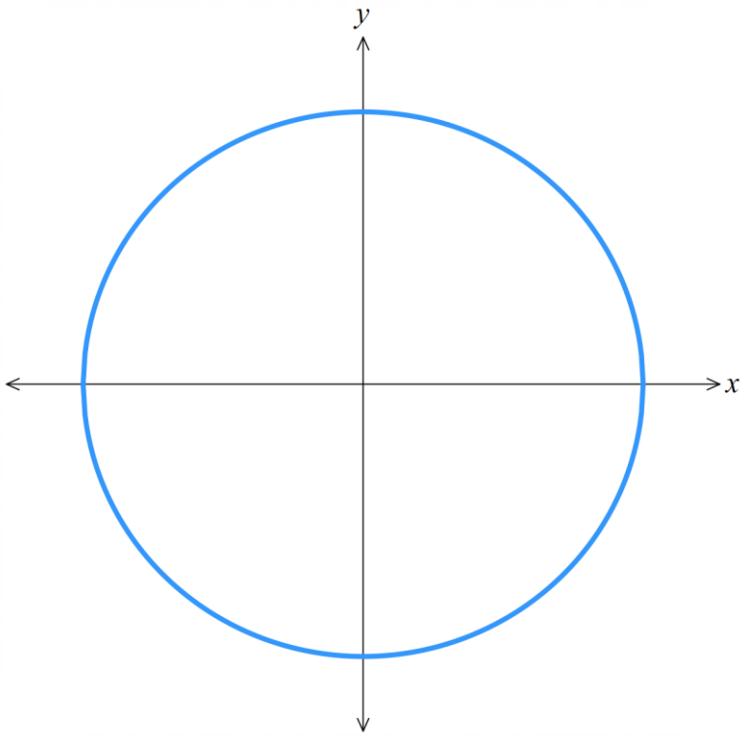


$y = -\frac{3}{x}$

Circles at the Origin

Investigation Equation of a Circle

Consider the following circle with radius r centred at the origin $(0,0)$



1. Choose a point on the circumference. Label this $P(x, y)$

2. Draw a line segment from the centre to the point P.
   What is the length of this line and what is it called? ……

3. Construct a right-angled triangle using the radius as the hypotenuse.
   Label the lengths of the sides.

4. Write an equation relating the three sides of the right-angled triangle.

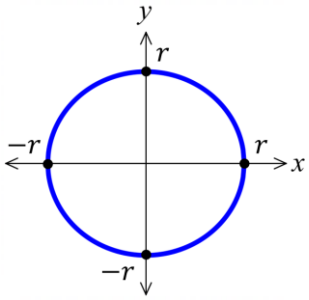
   ……

   This is the equation of a circle.

ⓘ Equation of a Circle Centred at the Origin

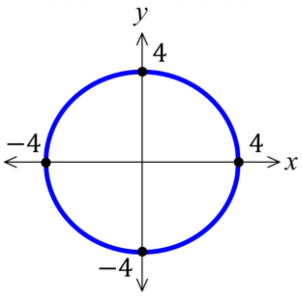
$$x^2 + y^2 = r^2$$

↑ radius r

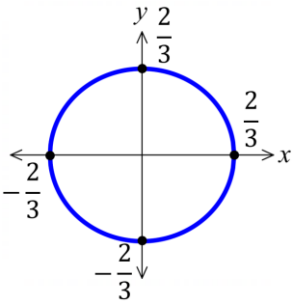


ⓘ **Sketch** the graph of a circle centred at the origin

Sketch $x^2 + y^2 = 16$

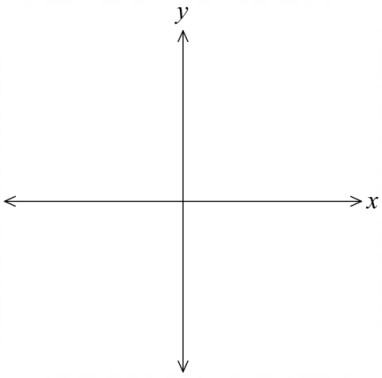


Sketch $x^2 + y^2 = \frac{4}{9}$

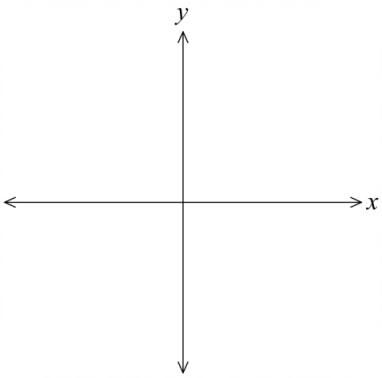


Sketch these circles

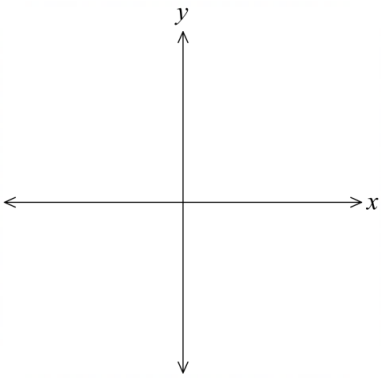
a $x^2 + y^2 = 36$



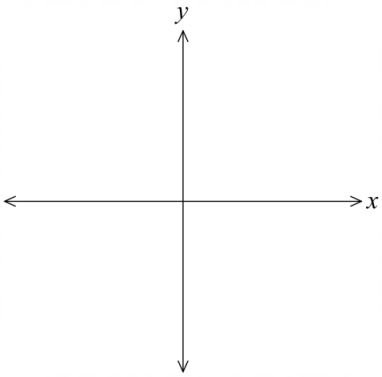
b $x^2 + y^2 = 37$



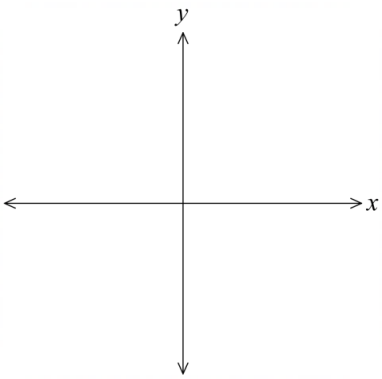
c $x^2 + y^2 = \frac{1}{4}$



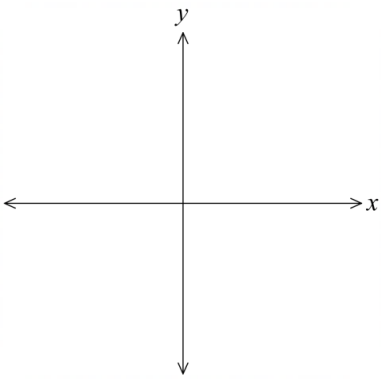
d $x^2 + y^2 = \frac{9}{4}$



e $x^2 + y^2 = 144$



f $x^2 + y^2 = 7^2$



1 Identify the radius of the circles with these equations:

a $x^2 + y^2 = 36$

6

b $x^2 + y^2 = 81$

9

c $x^2 + y^2 = 144$

12

d $x^2 + y^2 = 5$

$\sqrt{5}$

e $x^2 + y^2 = 12$

$2\sqrt{3}$

f $x^2 + y^2 = \frac{1}{4}$

$\frac{1}{2}$

2 Write the equation of a circle with centre at the origin and radius of:

a 4

$x^2 + y^2 = 16$

b 7

$x^2 + y^2 = 49$

c 100

$x^2 + y^2 = 10000$

d $\sqrt{6}$

$x^2 + y^2 = 6$

e $\sqrt{10}$

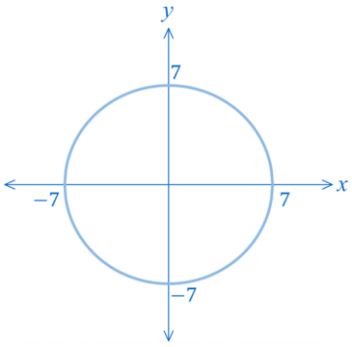
$x^2 + y^2 = 10$

f 1.1

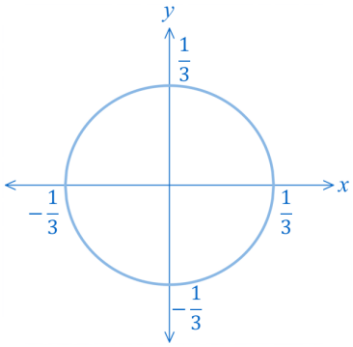
$x^2 + y^2 = 1.21$

3 Sketch these circles showing the centre and all intercepts.

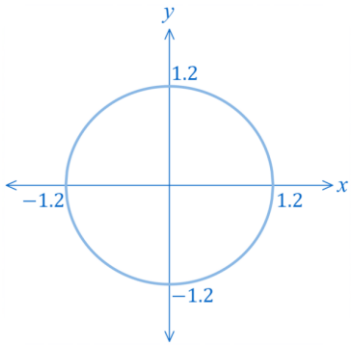
a $x^2 + y^2 = 49$



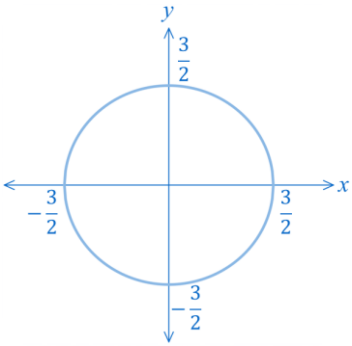
b $x^2 + y^2 = \frac{1}{9}$



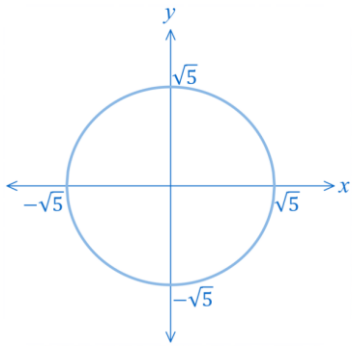
c $x^2 + y^2 = 1.44$



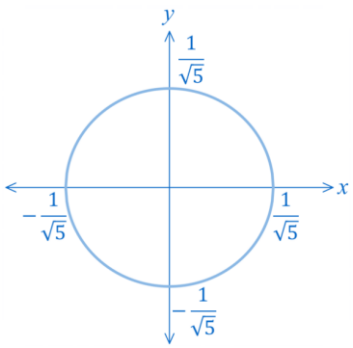
d $x^2 + y^2 = \frac{9}{4}$



e $x^2 + y^2 = 5$

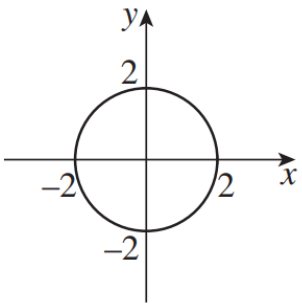


f $x^2 + y^2 = \frac{1}{5}$



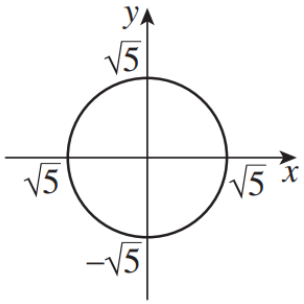
4 Determine the equation of each circle. They are all centred at the origin.

a



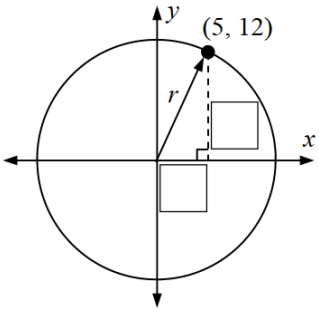
$x^2 + y^2 = 4$

b



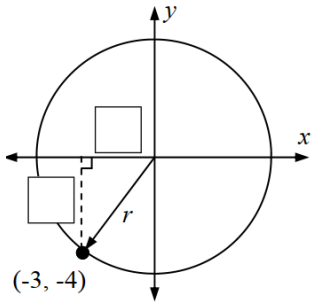
$x^2 + y^2 = 5$

c



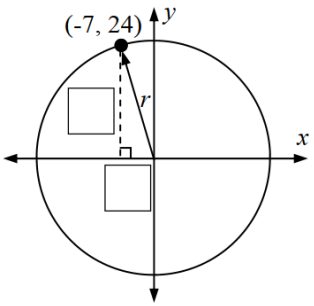
$x^2 + y^2 = 169$

d



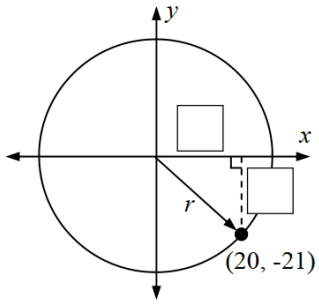
$x^2 + y^2 = 25$

e



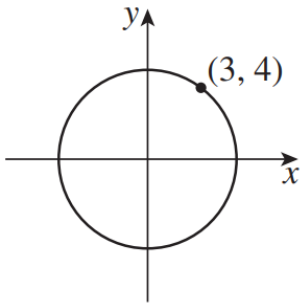
$x^2 + y^2 = 625$

f



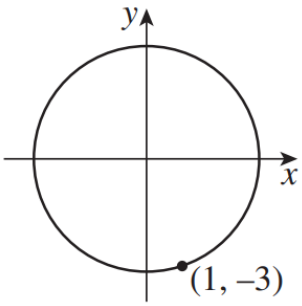
$x^2 + y^2 = 841$

g



$x^2 + y^2 = 25$

h



$x^2 + y^2 = 10$

5 A circle with centre $O(0,0)$ passes through $A(5,12)$.

a What is its radius?

13

b Therefore, what is its equation?

$x^2 + y^2 = 169$

6 Consider the four points $A(1,4), B\left(2,\sqrt{13}\right), C\left(3,2\sqrt{2}\right)$ and $D(4,1)$

a Find the distance of each point from the origin O.

$\sqrt{17}$

b Hence explain why the four points lie on a circle with centre the origin and state the equation of the circle.

Each point is equidistant from the origin, so it lies on a circle centred at the origin.

$x^2 + y^2 = 17$

c What is the radius, diameter, circumference and area of this circle?

$r = \sqrt{17}$, $d = 2\sqrt{17}$, $C = 2\pi\sqrt{17}$, $A = 17\pi$

7 Show, using substitution, that $E(-12,-5)$ lies on the circle centred at the origin with radius 13.

$x^2 + y^2 = 169$

$(-12)^2 + (-5)^2 = 144 + 25 = 169$

8 Consider points $A(-5,0), B(5,0)$, and $C(3,4)$.

a Show that the points all lie on the circle $x^2 + y^2 = 25$.

Using distance formula, distance from origin of all 3 points is 5

b Show that $AC \perp BC$.

$m_{AC} = \frac{1}{2}, m_{BC} = -2$

Since the gradients of AC and BC are negative reciprocals, they are perpendicular.

Circle Transformations

ⓘ General Equation of a Circle

centre (a, b)

$$(x-a)^2+(y-b)^2=r^2$$

radius r

ⓘ **Identify** Centre and radius from equation of a circle

$(x-3)^2+(y-4)^2=25$

Centre $(3,4)$, Radius 5

$(x-4)^2+(y+3)^2=36$

Centre $(4,-3)$, Radius 6

Identify the centre and radius of these circles

a $(x-3)^2+(y+4)^2=16$

Centre ..... Radius .....

b $(x+3)^2+y^2=25$

Centre ..... Radius .....

c $(x+2)^2+(y-50)^2=100$

Centre ..... Radius .....

d $(x+19)^2+\left(y+\frac{1}{2}\right)^2=50$

Centre ..... Radius .....

Find the equation of the circle for:

e Centre: $(3,9)$, radius 4

f Centre: $\left(-1,\frac{3}{4}\right)$, radius $\sqrt{13}$

Sketching a Translated Circle

centre (a, b)

$$(x-a)^2+(y-b)^2=r^2$$

radius r

1. Identify centre and radius.
2. Draw a quick sketch, using the radius to determine if it crosses the axes.
3. Find x and y intercepts, if any.

Example Sketch the graph of a circle not centred at the origin

Sketch $(x-1)^2+(y+2)^2=9$.

Centre $(1,-2)$ radius 3.

The circle will cross the axes because the radius is 3.

x intercepts let $y=0$

$(x-1)^2+(0+2)^2=9$

$(x-1)^2=5$

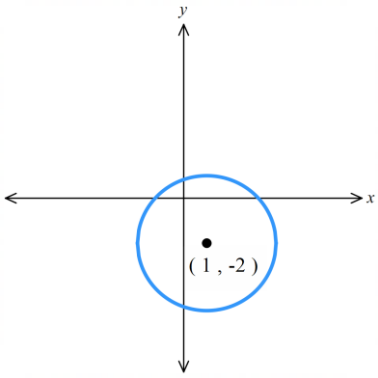
$x=1\pm\sqrt{5}$

y intercepts let $x=0$

$(0-1)^2+(y+2)^2=9$

$(y+2)^2=8$

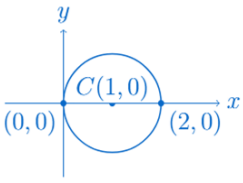
$y=-2\pm 2\sqrt{2}$



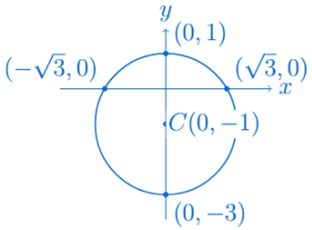
Guided Practice

Sketch, and find the intercepts, if any.

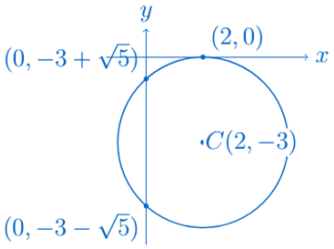
a $(x-1)^2+y^2=1$



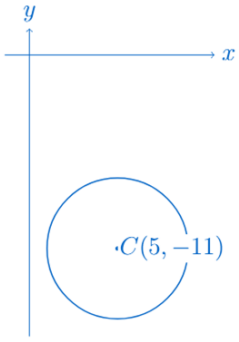
b $x^2+(y+1)^2=4$



c $(x-2)^2+(y+3)^2=9$



d $(x-5)^2+(y+11)^2=16$



1 Choose the word *left, right, up* or *down* to complete each sentence.

a Compared to $x^2 + y^2 = r^2$, $(x-1)^2 + y^2 = r^2$ is translated ..... by 1 unit.

right

b Compared to $x^2 + y^2 = r^2$, $(x+3)^2 + y^2 = r^2$ is translated .....

Left by 3 units

c Compared to $x^2 + y^2 = r^2$, $x^2 + (y+3)^2 = r^2$ is translated .....

Down by 3 units

d Compared to $x^2 + y^2 = r^2$, $x^2 + (y-2)^2 = r^2$ is translated .....

Up by 2 units

e Compared to $x^2 + y^2 = r^2$, $(x+2)^2 + (y-1)^2 = r^2$ is translated

.....

Left by 2, up by 1

f Compared to $x^2 + y^2 = r^2$, $(x+5)^2 + (y+3)^2 = r^2$ is translated

.....

Left by 5, down by 3

g Compared to $x^2 + y^2 = r^2$, $(x-4)^2 + (y+2)^2 = r^2$ is translated

.....

Right by 4, down by 2

2 Write the equation of the circle with:

a centre $(-1,5)$, radius 4

$(x+1)^2 + (y-5)^2 = 16$

b centre $(3,-5)$, radius $\sqrt{7}$

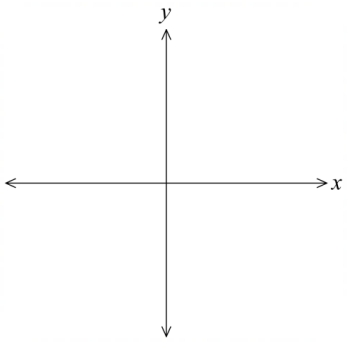
$(x-3)^2 + (y+5)^2 = 7$

c $(8,11)$, radius $2\sqrt{5}$

$(x-8)^2 + (y-11)^2 = 20$

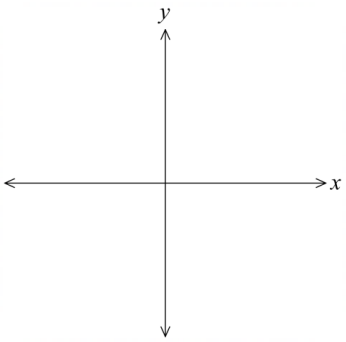
3 Sketch the following circles.
Label the coordinates of the centre and intercepts, if any. Check your sketch using DESMOS.

a $(x-1)^2+y^2=1$



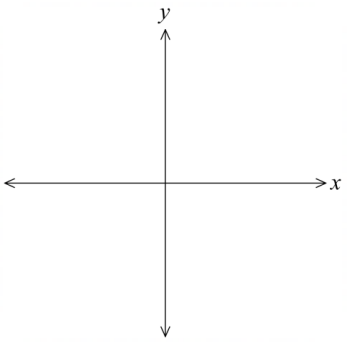
Centre: (1,0)
x-intercepts:$x=0,2$
y-intercepts:$y=0$

b $(x+1)^2+y^2=9$



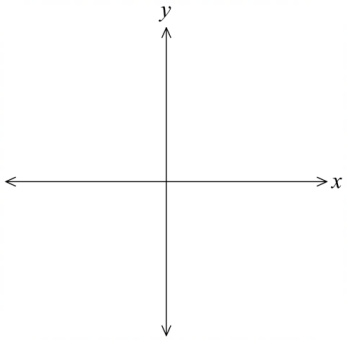
Centre: (−1,0)
x-intercepts:$x=-4,2$
y-intercepts:$y=\pm 2\sqrt{2}$

c $x^2+(y-4)^2=36$



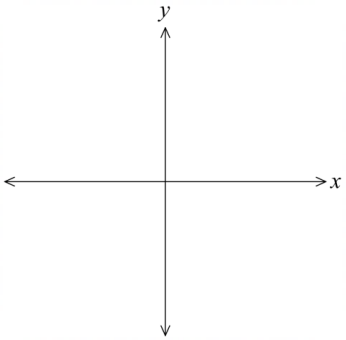
Centre: (0,4)
x-intercepts:$x=\pm 2\sqrt{5}$
y-intercepts: $-2,10$

d $(x-3)^2+(y+1)^2=1$



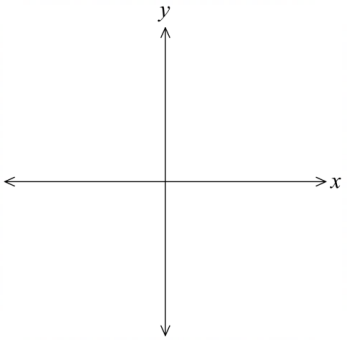
Centre: (3,−1)
x-intercepts:$x=3$
y-intercepts: none

e $(x+2)^2+(y-3)^2=4$



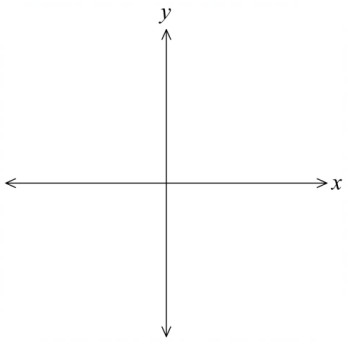
Centre: (−2,3)
x-intercepts: none
y-intercepts:3

f $(x-1)^2+(y+3)^2=25$



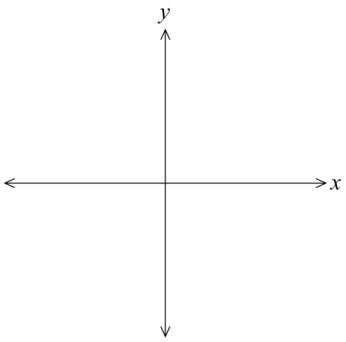
Centre: (1,−3)
x-intercepts:$x=-3,5$
y-intercepts:$y=-3\pm 2\sqrt{6}$

g $(x+3)^2+(y-2)^2=25$



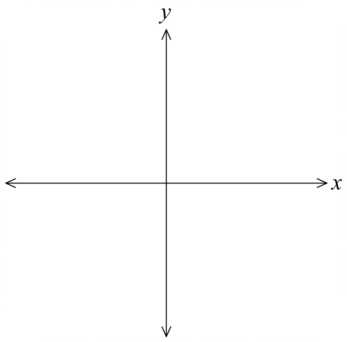
Centre: $(-3,2)$
x-intercepts: $x=-3\pm\sqrt{21}$
y-intercepts: $y=-2,6$

h $(x+2)^2+(y-1)^2=9$



Centre: $(-2,1)$
x-intercepts: $x=-2\pm2\sqrt{2}$
y-intercepts: $y=1\pm\sqrt{5}$

i $(x-2)^2+(y-5)^2=64$



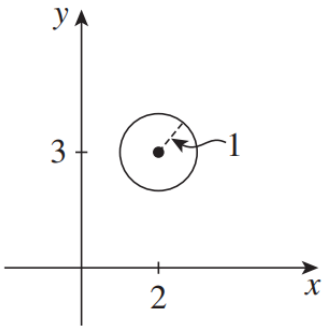
Centre: $(2,5)$
x-intercepts: $x=2\pm\sqrt{39}$
y-intercepts: $y=5\pm2\sqrt{15}$

4 A circle with centre $B(4,5)$ passes through the origin. What is its equation?

$(x-4)^2+(y-5)^2=41$

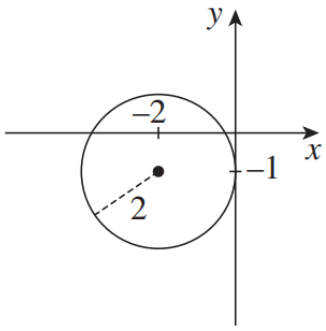
5 Determine the equation of each circle, using Pythagoras if necessary to find the radius.

a



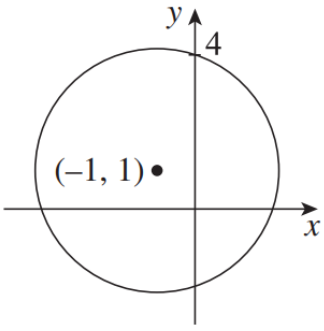
$(x-2)^2+(y-3)^2=1$

b



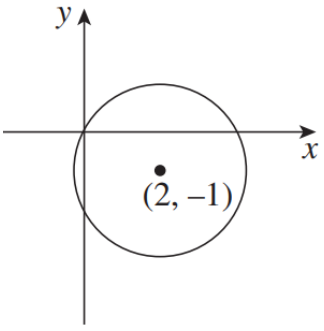
$(x+2)^2+(y+1)^2=4$

c



$(x+1)^2+(y-1)^2=10$

d



$(x-2)^2+(y+1)^2=5$

6 Find the equation of a circle with centre $(5, -2)$ passing through $(-1,1)$.

Distance between the centre and the point is $\sqrt{45}$, so the radius is $\sqrt{45}$.

$(x-5)^2 + (y+2)^2 = 45$

7 A circle has centre $(4, -3)$ and one of the x-intercepts is at $(-\sqrt{7} + 4, 0)$.
Determine the equation of the circle.

Distance between the centre and the intercept is 4, so the radius is 4.

$(x-4)^2 + (y+3)^2 = 16$

8 The interval joining the points $A(3, -1)$ and $B(-7,5)$ is a diameter of a circle.

a Find the midpoint of AB.

$(-2,2)$

b Explain why the midpoint of AB is the centre of the circle.

The diameter connects two points on the circumference, passing through the centre.

Therefore, the midpoint of the diameter is the centre of the circle.

c Find the length of the radius.

$\sqrt{34}$

d Hence state the equation of the circle.

$(x+2)^2 + (y-2)^2 = 34$

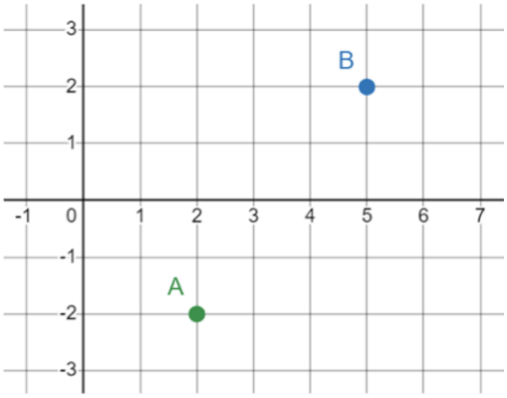
9 Find the centre of the circle with diameter CD, where $C = (2, 1)$ and $D = (8, -7)$.

$(5, -3)$

10 Find the equation of a circle with diameter KL, where $K(5,7)$ and $L(-9,3)$

$(x + 2)^2 + (y - 5)^2 = 53$

11 Find the possible equations of a circle with radius AB.



Circle with centre at A:

$(x - 2)^2 + (y + 2)^2 = 25$

Circle with centre at B:

$(x - 5)^2 + (y - 2)^2 = 25$

Completing the Square for a Circle

Completing the Square to Find the Centre and Radius

Some questions require you to **complete the square** for the x and y terms to write in the form

$$(x-a)^2+(y-b)^2=r^2$$

Example Sketch the graph of a circle by completing the square

$$x^2+2x+y^2-6y-6=0$$

$$x^2+2x+\mathbf{1}-\mathbf{1}+y^2-6y+\mathbf{9}-\mathbf{9}-6=0$$

$$(x^2+2x+1)-1+(y^2-6y+9)-9-6=0$$

$$(x+1)^2+(y-3)^2-16=0$$

$$(x+1)^2+(y-3)^2=16$$

Centre: $(-1,3)$, radius 4

Guided Practice

Find the centre and radius by completing the square first.

a $x^2-4x+y^2=0$

$(2,0)$ $r=2$

b $x^2+6x+y^2-8y=0$

$(-3,4)$ $r=5$

c $x^2+y^2-10x+12y=12=0$

$(5,-6)$ $r=7$

d $4x^2+4y^2-12x+20y-7=0$

$\left(\frac{3}{2},-\frac{5}{2}\right)$ $r=\frac{\sqrt{41}}{2}$

1 Determine the centre and radius of these circles by completing the square, if necessary.

a $(x+1)^2+y^2=4$

Centre: $(-1,0)$ Radius: 2

b $(x-1)^2+(y-2)^2=1$

Centre: $(1,2)$ Radius: 1

c $x^2-10x+y^2=-16$

Centre: $(5,0)$ Radius: 3

d $x^2+y^2+14x=33$

Centre: $(-7,0)$ Radius: $\sqrt{82}$

e $x^2-2x+y^2-4y-4=0$

Centre: $(1,2)$ Radius: 3

f $x^2+6x+y^2-8y=0$

Centre: $(-3,4)$ Radius: 5

g $x^2-10x+y^2+8y+32=0$

Centre: $(5,-4)$ Radius: 3

h $x^2+14x+14+y^2-2y=0$

Centre: $(-7,1)$ Radius: 6

i $x^2+4x+y^2-2y+1=0$

Centre: $(-2,1)$ Radius: 2

j $x^2-6x+y^2-4y-3=0$

Centre: $(3,2)$ Radius: 4

2

a Complete the square for $x^2+4x+y^2-6y+15=0$.

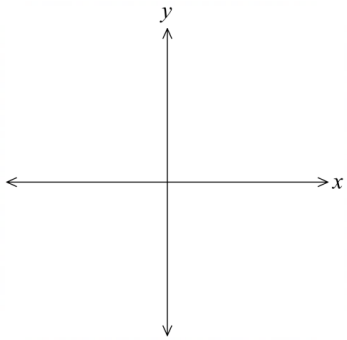
$(x+2)^2+(y-3)^2=-2$

b Explain why this is not an equation of a circle.

The radius cannot be negative.

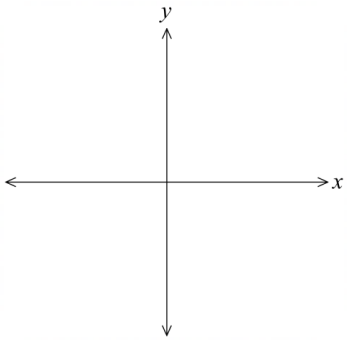
3 Complete the square to sketch these circles. Show the centre, radius, and intercepts.

a $x^2 - 2x + y^2 = 0$



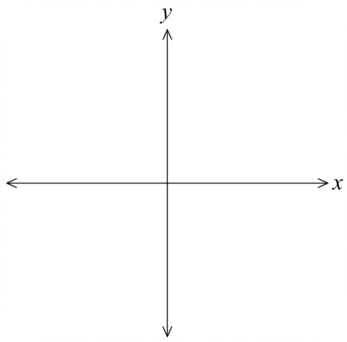
Centre: (1,0)
Radius: 1
x-intercepts: $x = 0, 2$
y-intercepts: $y = 0$

b $x^2 + 6x + y^2 = 0$



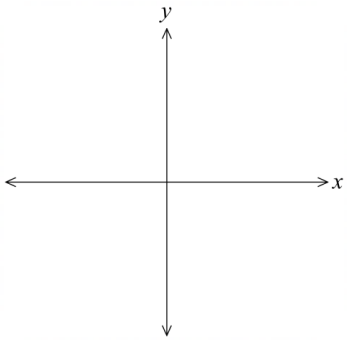
Centre: (–3,0)
Radius: 3
x-intercepts: $x = -6, 0$
y-intercepts: $y = 0$

c $x^2 + 4x + y^2 - 2y + 1 = 0$



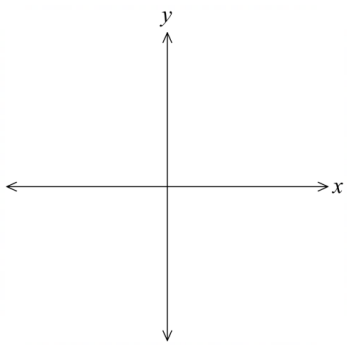
Centre: (–2,1)
Radius: 2
x-intercepts: $x = -2 \pm \sqrt{3}$
y-intercepts: $y = 1$

d $x^2 + 8x + y^2 + 10y + 5 = 0$



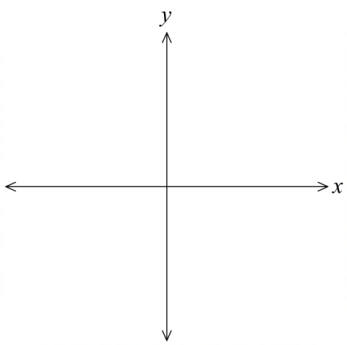
Centre: (–4,–5)
Radius: 6
x-intercepts: $x = -4 \pm \sqrt{11}$
y-intercepts: $y = -5 \pm 2\sqrt{5}$

e $x^2 - 6x + y^2 - 4y - 3 = 0$



Centre: (3,2)
Radius: 4
x-intercepts: $x = 3 \pm 2\sqrt{3}$
y-intercepts: $y = 2 \pm \sqrt{7}$

f $x^2 - 2x + y^2 + 6y - 5 = 0$



Centre: (1,–3)
Radius: $\sqrt{15}$
x-intercepts: $x = 1 \pm \sqrt{6}$
y-intercepts: $y = -3 \pm \sqrt{14}$

4 For what values of k would the graph of $x^2 + 6x + y^2 + k = 0$ not exist?

By CTS, $(x+3)^2 + y^2 = 9 - k$

For circle to not exist, $9 - k < 0$

$\therefore k > 9$

5 A naval ship is at coordinates P and equipped with a radar scanner that can detect objects inside the circle $x^2 + y^2 - 80x - 40y = 500$, where radius is measured in kilometres.

a Find the coordinates of P.

By CTS, $(x-40)^2 + (y-20)^2 = 2500$

P is the centre of the circle, at $(40,20)$

b Another ship is at $Q(60, -20)$ and can detect objects within a radius of 40 km.

i Will ship Q be able to detect P?

Distance between PQ is $\sqrt{2000} \approx 44.7$ km. Since Q has a radar range of 40 km, no.

ii Will ship P be able to detect Q?

P's radar has a range of 50 km, so yes.

6 Consider the points $A(1,3)$, $B(3,9)$ and let $P(x,y)$ be a point.

a Find an expression for the gradients of PA and PB.

$$m_{PA} = \frac{y-3}{x-1}, \quad m_{PB} = \frac{y-9}{x-3}$$

b Suppose $\angle APB = 90°$. Show that $x^2 - 4x + y^2 - 12y + 30 = 0$.

$$\begin{aligned} m_{PA} \times m_{PB} &= -1 \\ \frac{y-3}{x-1} \times \frac{y-9}{x-3} &= -1 \\ \frac{(y-3)(y-9)}{(x-1)(x-3)} &= -1 \\ (y-3)(y-9) &= -(x-1)(x-3) \\ y^2 - 12y + 27 &= -x^2 + 4x - 3 \\ x^2 - 4x + y^2 - 12y + 30 &= 0 \end{aligned}$$

c By completing the square, show that this is the equation of a circle centred at $(2,6)$ with radius $\sqrt{10}$.

d How is the point P related to the circle? A sketch using DESMOS may help.
https://www.desmos.com/geometry/opfcwyddok

We have proved that if we draw any point P and connect it to both A and B such that it forms a right angle at $\angle APB$, the point P must lie on a circle centred at the midpoint of AB with radius $\sqrt{10}$.

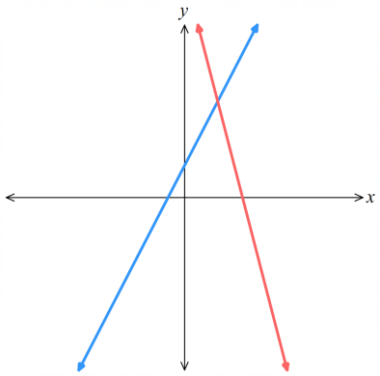
This is related to the circle theorem "The angle at the circumference in a semicircle is 90°"

Points of Intersection of a Parabola and a Line

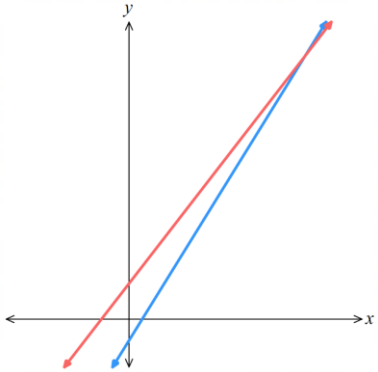
Review

☐ Recall finding points of intersection of two lines

a $y = 2x + 1$ and $y = -4x + 7$



b $y = 5x - 3$ and $y = 4x + 5$



☐ Solve non-linear simultaneous equations

a Solve
$y = x - 1$ and $y = x^2 - 3$

$x = 2, y = 1$
$x = -1, y = -2$

b Solve
$y = x^2 - 3x + 7$ and $y = 2x + 3$

$x = 1, y = 5$
$x = 4, y = 11$

Finding the points of Intersection of a Parabola and a Line

When you have two graphs: $y = ax^2 + bx + c$ and $y = mx + d$. The **point of intersection** lies on both curves, so is the **simultaneous solution** to the equations.

1. Set the two y values to be equal $y_1 = y_2$
2. Solve for x by rearranging the resulting quadratic equation.
3. Solve for y.

Example Find points of intersection of a parabola and a line

Find the points of intersection of $y = x^2 + 4x + 4$ and $y = 16$.

1. $x^2 + 4x + 4 = 16$
2. $x^2 + 4x - 12 = 0$

$(x + 6)(x - 2) = 0$

$x = -6$ or 2

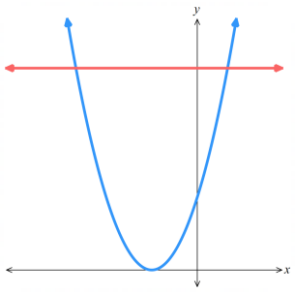
3. when $x = -6$, $y = 16$
   when $x = 2$, $y = 16$

so, points of intersection are $(-6,16)$ and $(2,16)$

Guided Practice

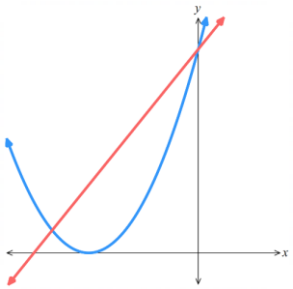
Find the points of intersection of:

a $y = x^2 + 6x + 9$ and $y = 25$



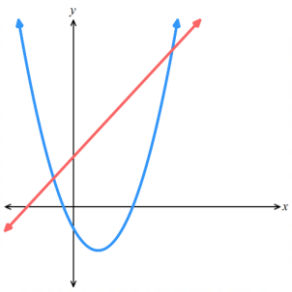
$(-8,25)$ and $(2,25)$

b $y = x^2 + 6x + 9$ and $y = 2x + 9$



$(-4,1)$ and $(0,9)$

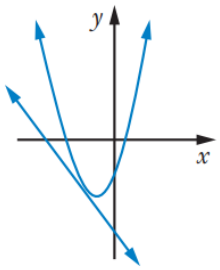
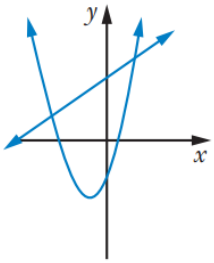
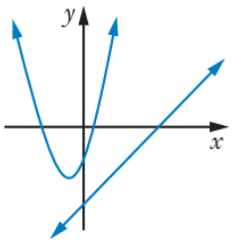
c $y = 2x^2 - 5x - 3$ and $y = 3x + 7$



$(-1,4)$ and $(5,22)$

ⓘ The Discriminant and Points of Intersection

A parabola intersects a straight line at **zero**, **one**, or **two** points.
The discriminant of the resulting equation after setting the two equations equal reveals the number of points of intersection.

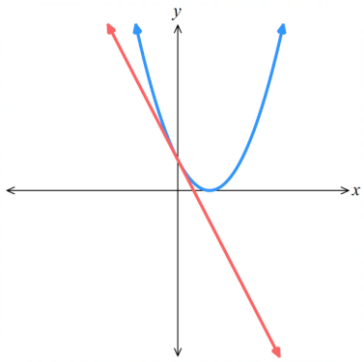
$\Delta = 0$	$\Delta > 0$	$\Delta < 0$
1 intersection point	Two intersection points	No intersection points
		
the line is **tangent** to the parabola		

ⓘ **Investigation** The discriminant and points of intersection

Consider the two graphs of $y = x^2 - 2x + 1$ and $y = -2x + 1$.

The graph shows that there is only one point of intersection.

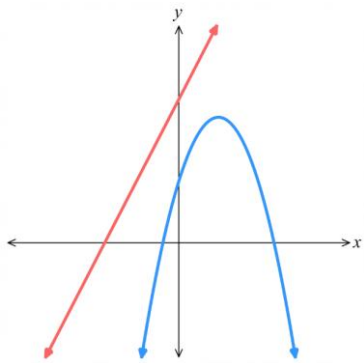
Show that $\Delta = 0$ and then find the point of intersection.



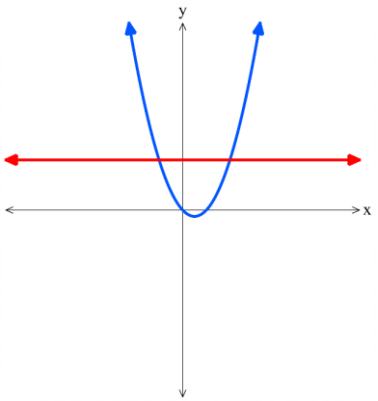
Consider the two graphs of $y = -2x^2 + 5x + 3$ and $y = 3x + 7$.

The graph shows no points of intersection.

Find the value of Δ and confirm this is true.



1 The graphs of $y = x^2 - x$ and $y = 2$ are shown. Solve the two equations simultaneously for x.



Equate:

$x^2 - x = 2$

Rearrange:

Solve for x:

$(-1,2)$ and $(2,2)$

Do your solutions roughly match what is shown on the graph? ……

yes

2 The graph of a parabola $y = x^2 - x - 1$ is shown.

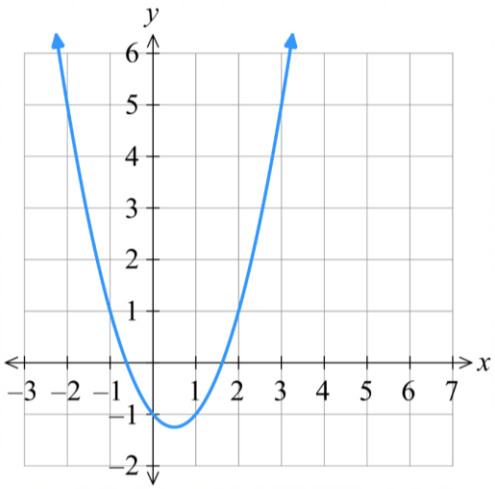
a Sketch the line $y = x + 2$ on this graph.

b Use the grid to find the points of intersection.

……

c Find the points of intersection algebraically:

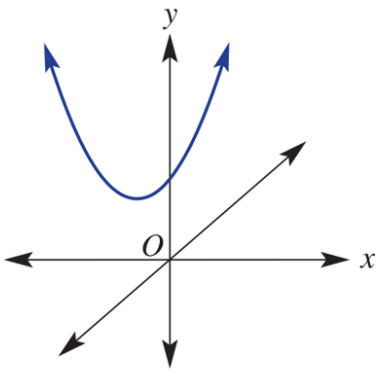
$x^2 - x - 1 = x + 2$



$(-1,1)$ and $(3,5)$

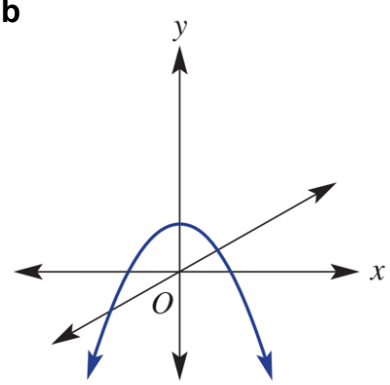
3 What would be the value of the discriminant $\Delta = b^2 - 4ac$ from solving the following equations simultaneously? Answer with $\Delta < 0$, $\Delta = 0$, or $\Delta > 0$.

a



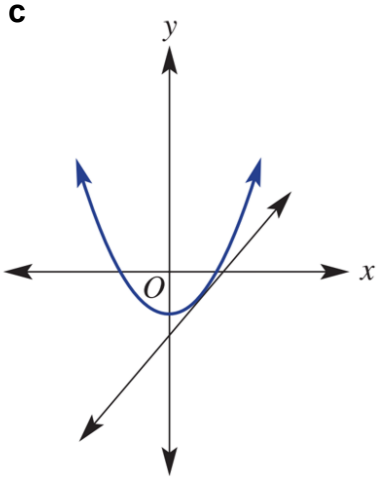
$\Delta < 0$

b



$\Delta > 0$

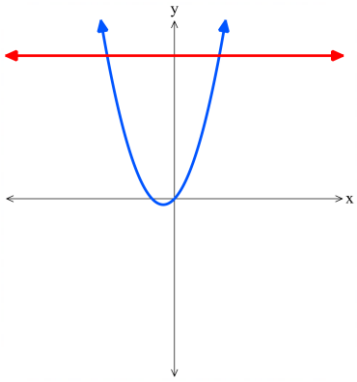
c



$\Delta = 0$

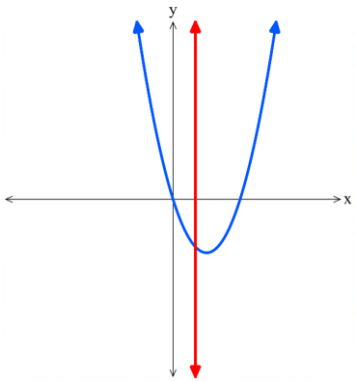
4 Find the points of intersection for these lines and parabolas.

a $y = x^2 + x$ and $y = 6$



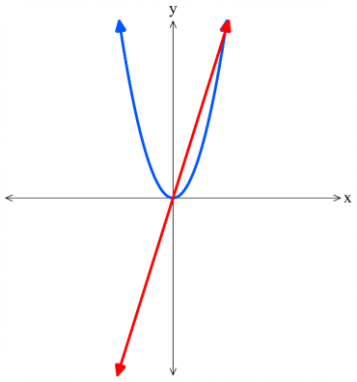
$(-3,6)$ and $(2,6)$

b $y = x^2 - 3x$ and $x = 1$



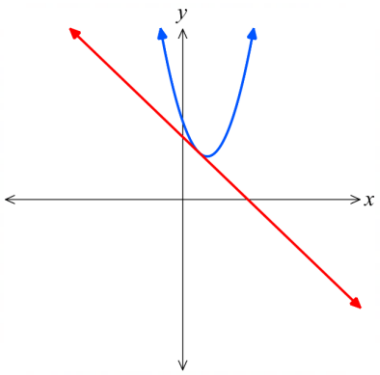
$(1,-2)$

c $y = x^2$ and $y = 3x$



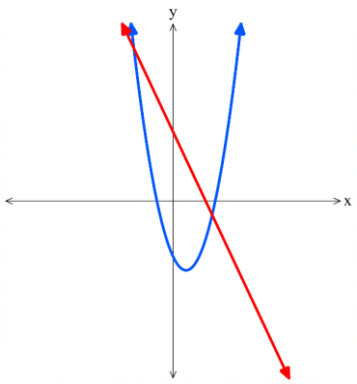
$(0,0)$ and $(3,9)$

d $y = x^2 - 3x + 5$ and $y = -x + 4$



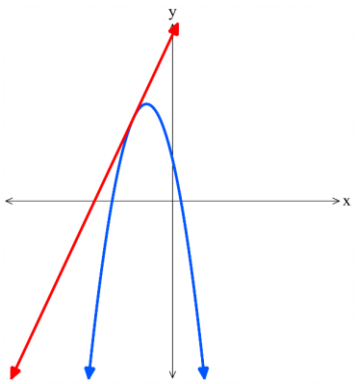
$(1,3)$

e $y = x^2 - 2x - 4$ and $y = -2x + 5$



$(-3,11)$ and $(3,-1)$

f $y = -x^2 - 4x + 3$ and $y = 2x + 12$



$(-3,6)$

5 Solve the following simultaneous equations, making use of the quadratic formula. Leave answers in exact surd form. Check your answer using DESMOS.

a Find the point of intersection of $y = 2x^2 + 3x + 6$ and $y = x + 4$

No point of intersection.

b Find the point of intersection of $y = x^2 + 2x - 5$ and $y = x$

$\left(\frac{-1-\sqrt{21}}{2}, \frac{-1-\sqrt{21}}{2}\right)$ and $\left(\frac{-1+\sqrt{21}}{2}, \frac{-1+\sqrt{21}}{2}\right)$

6 A member of an indoor cricket team, playing a match in a gymnasium, hits a ball that follows a path given by $y = -0.1x^2 + 2x + 1$, where y is the height above ground, in metres, and x is the horizontal distance travelled by the ball. The ceiling of the gymnasium is 10.6 metres high. Determine whether the ball will intersect with the ceiling.

Set up equation $10.6 = -0.1x^2 + 2x + 1$

Yes, as $\Delta > 0$.

7

a Use the discriminant to show that the line $y = 2x + 1$ does not intersect the parabola $y = x^2 + 3$.

$\Delta = -4$, so no solutions.

b Determine for which values of k the line $y = 2x + k$ intersects the parabola $y = x^2 + 3$.

$2x + k = x^2 + 3$

$x^2 - 2x + 3 - k = 0$

$\Delta = 4 - 4(1)(3 - k)$

$= 4 - 12 + 4k$

$= 4k - 8$

$4k - 8 \geq 0$

$k \geq 2$

8 The line $y = mx$ is a **tangent** to the parabola $y = x^2 - 2x + 4$
(i.e the line touches the parabola at just one point).

a Find the possible values of m.

Set up equation $mx = x^2 - 2x + 4$

$x^2 + (-2 - m)x + 4 = 0$

Tangent means 1 intersection point. $b^2 - 4ac = 0$

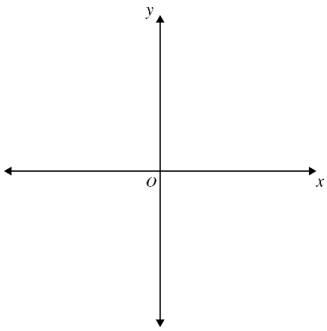
$(-2 - m)^2 - 4(1)(4) = 0$

$m^2 + 4m - 12 = 0$

$(m + 6)(m - 2) = 0$

$m = 2$ or -6

b Explain why there are two possible values of m.
A diagram may help (use DESMOS to sketch $y = x^2 - 2x + 4$ and $y = mx$).



There are two ways to draw a line that is tangent to $y = x^2 - 2x + 4$,
one with gradient -6 and the other with gradient 2

c If the value of m is changed so that the line now intersects the parabola in two places, what is the set of possible values of m?

$\Delta > 0$

$(m + 6)(m - 2) > 0$

By sketching the parabola $y = (x + 6)(x - 2)$ and solving the inequality, $m < -6, m > 2$

9 The line $y = mx - 4$ is tangent to the parabola $y = x^2 - 2x + 5$. Find the values of m.

4 or -8

10 Find the values of p for which $y = px - 3p - 5$ is tangent to $x^2 = 8y$.

$p = \frac{5}{2}$ or $p = -1$

Points of Intersection of Two Curves

ⓘ Finding the Points of Intersection of Any Two Graphs

For any two graphs,

1. Solve simultaneously to find the x-coordinate(s) of the point of intersection.

2. Find the y-coordinate(s).

Example Find points of intersection of a line and a hyperbola

Find the points of intersection of the line $y = 4x$ and the hyperbola $y = \frac{1}{x}$

1. Solve for x.

$4x = \frac{1}{x}$

$4x^2 = 1$

$x^2 = \frac{1}{4}$

$x = \pm\frac{1}{2}$

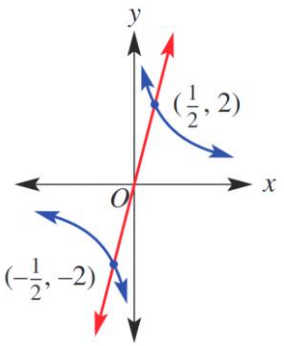
2. Find y values.

Substitute x into (1) to find y values:

When $x = \frac{1}{2}$ $\quad y = 4\left(\frac{1}{2}\right) = 2$

When $x = \frac{-1}{2}$ $\quad y = 4\left(\frac{-1}{2}\right) = -2$

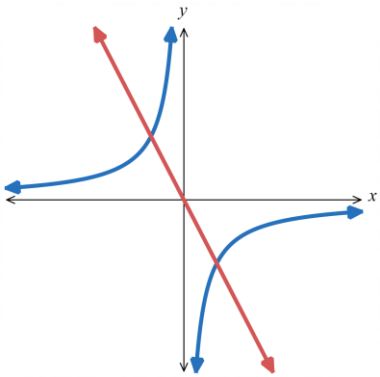
$\therefore \left(\frac{1}{2}, 2\right)$ and $\left(-\frac{1}{2}, -2\right)$



Guided Practice

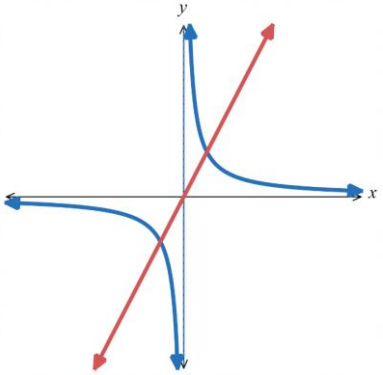
Find the point of intersection of these graphs.

a $y = -\frac{2}{x}$ and $y = -2x$



$(-1,2)$ and $(1,-2)$

b $y = \frac{1}{x}$ and $y = 2x$



$\left(-\frac{1}{\sqrt{2}}, -\sqrt{2}\right)$ and $\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$

Example Find points of intersection of a line and a circle

Find the points of intersection of the line $y = 2x$ and the circle $y^2 + x^2 = 4$

1. Solve simultaneously

$y = 2x$ (1)

$y^2 + x^2 = 4$ (2)

Sub (1) into (2)

$(2x)^2 + x^2 = 4$

$5x^2 = 4$

$x^2 = \frac{4}{5}$

$x = \pm\frac{2}{\sqrt{5}}$

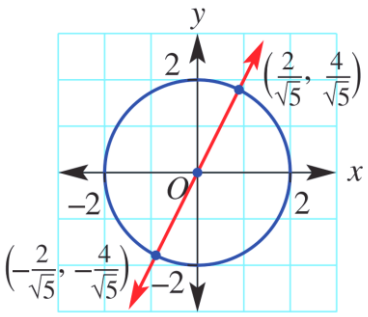
3. Find y values.

Substitute into (1) to find y values:

When $x = \frac{2}{\sqrt{5}}$ $\quad y = 2\left(\frac{2}{\sqrt{5}}\right) = \frac{4}{\sqrt{5}}$

When $x = \frac{-2}{\sqrt{5}}$ $\quad y = 2\left(\frac{-2}{\sqrt{5}}\right) = \frac{-4}{\sqrt{5}}$

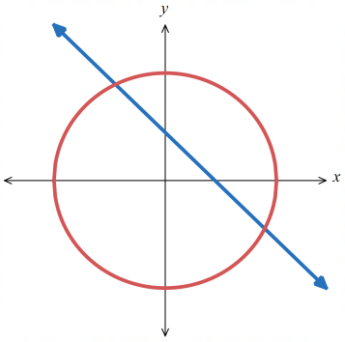
$\therefore \left(\frac{2}{\sqrt{5}}, \frac{4}{\sqrt{5}}\right)$ and $\left(\frac{-2}{\sqrt{5}}, \frac{-4}{\sqrt{5}}\right)$



Guided Practice

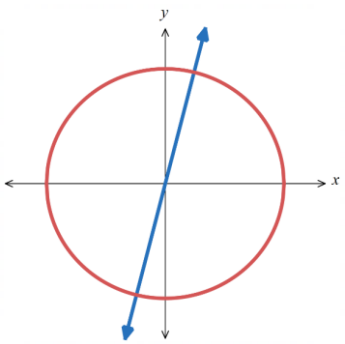
Find the point of intersection of these graphs.

a $y = -x + 1$ and
$x^2 + y^2 = 5$



$(-1,2)$ and $(2,-1)$

b $x^2 + y^2 = 16$ and
$y = 4x$



$\left(-\frac{4}{\sqrt{17}}, -\frac{16}{\sqrt{17}}\right)$ and $\left(\frac{4}{\sqrt{17}}, \frac{16}{\sqrt{17}}\right)$

1 Find the point of intersection of these graphs. You will need to rearrange before equating.

a $y = x^2$ and $x = \frac{y}{2}$

$(0,0)$ and $(2,4)$

b $y + 4x^2 + x - 6 = 0$ and $y - 3x = 7$

$(-0.5, 5.5)$

c $y = x^2 + 3x - 1$ and $3y = x - 3$

$\left(-\frac{8}{3}, -\frac{17}{9}\right)$ and $(0, -1)$

d $y - 7 = (x - 2)^2$ and $y + x = 9$

$(1,8)$ and $(2,7)$

e $xy = 1$ and $y = x - 1$

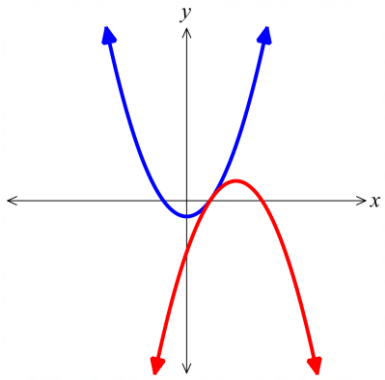
$\left(\frac{1-\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}\right)$ and $\left(\frac{1+\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2}\right)$

f $x^2 + y^2 = 4$ and $2y + x = 0$

$\left(\frac{-4}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$ and $\left(\frac{4}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right)$

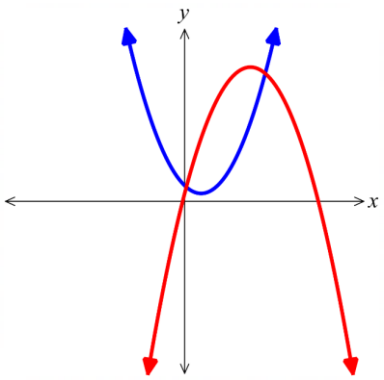
2 Find the point of intersection for these functions.
A sketch has been provided to help you interpret your answer.

a $y = x^2 - 2$
$y = -x^2 + 6x - 6.5$



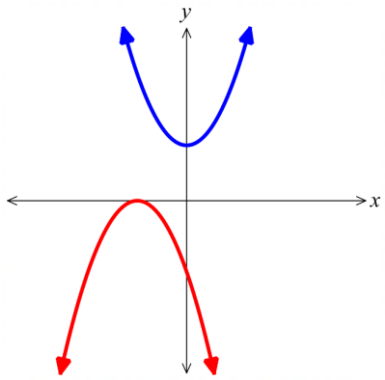
(1.5, 0.25)

b $y = x^2 - 2x + 2$
$y = -x^2 + 8x + 1$



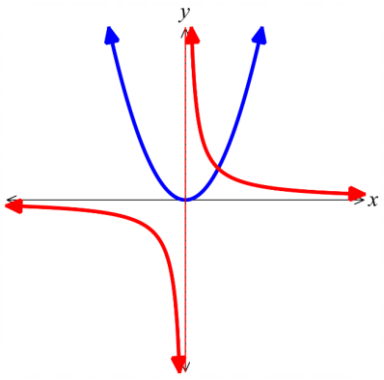
$\left(\frac{5-\sqrt{23}}{2}, 9-\frac{3\sqrt{23}}{2}\right)$ and $\left(\frac{5+\sqrt{23}}{2}, 9+\frac{3\sqrt{23}}{2}\right)$

c $y = x^2 + 7$
$y = -x^2 + 6x - 9$



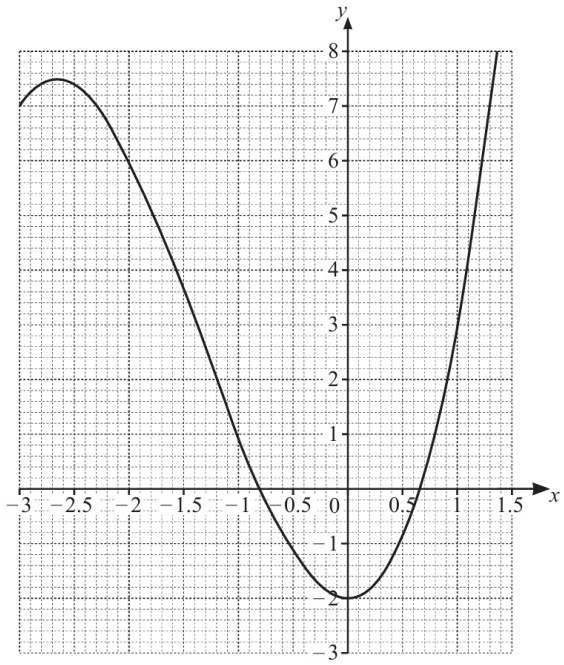
You will end up with a quadratic where $\Delta < 0$,
no points of intersection

d $y = x^2$
$y = \frac{8}{x}$



(2,4)

3 The diagram shows the graph of $y = x^3 + 4x^2 - 2$ for $-3 \leq x \leq 1.5$.
By drawing a suitable straight line, solve the equation $x^3 + 4x^2 - 2 = 2x$ for $-3 \leq x \leq 1.5$.
Give your answer as estimates to 2 d.p.



Sketch the line $y = 2x$ on the same axis.

The x-coordinates of the points of intersection are the solution.

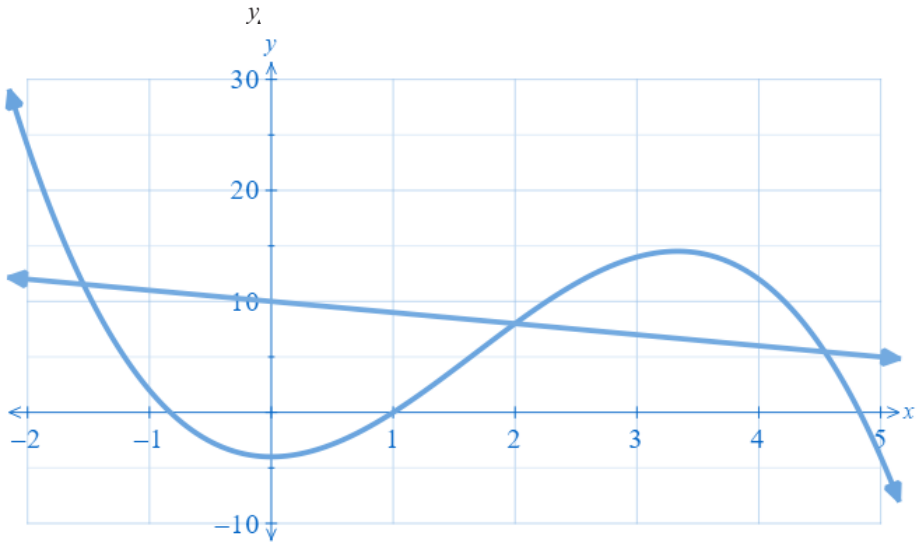
$x \approx -0.52$ and 0.88

4 The table shows some values for $y = 5x^2 - x^3 - 4$

x	-2	-1	0	1	2	3	4	5
y	24		-4		8	14		-4

a Complete the table.

b Sketch the graph for $-2 \leq x \leq 5$.



c By drawing a suitable straight line on the grid, solve the equation $x^3 - 5x^2 - x + 14 = 0$

This is equivalent to solving $5x^2 - x^3 - 4 = 10 - x$

Draw the line $y = 10 - x$ and read off the points of intersection

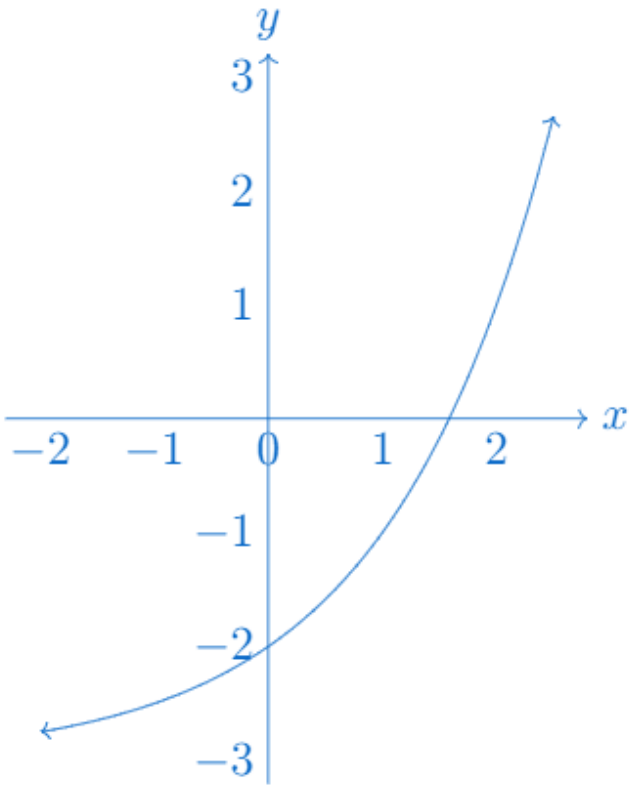
$x \approx -1, 5, 2, 4.5$

5 The table shows some values for $y = 2^x - 3$

a Complete the table.

x	-2	-1	0	0.5	1	1.5	2	2.5
y	-2.75			-1.58		-0.17	1	2.66

b On the grid, sketch the graph $y = 2^x - 3$ for $-2 \leq x \leq 2.5$.



c Use your graph to solve $2^x - 3 = 2$.

Draw a horizontal line $y = 2$ and read off points of intersection

$x \approx 2.3$

d By drawing a suitable straight line, solve $2^x - x - 1.5 = 0$.

This is equivalent to solving $2^x - 3 = x - 1.5$

Sketch $y = x - 1.5$ and read off x values of points of intersection

$x = -1$ and ≈ 1.66

6 Show that $y = -2x + 5$ is tangent to the circle $x^2 + y^2 = 5$.

Equate to find points of intersection:

$x^2 + (-2x + 5)^2 = 5$

$x^2 + 4x^2 - 20x + 25 = 5$

$5x^2 - 20x + 20 = 0$

$\Delta = b^2 - 4ac = (-20)^2 - 4(5)(20) = 0$

$\therefore$ there is 1 point of intersection, hence the line is tangent to the circle.

7 Show that $3x + y = 6$ is tangent to the circle $x^2 + y^2 + 4x + 16y + 28 = 0$.

Same method as above.

$x^2 + (-3x + 6)^2 + 4x + 16(-3x + 6) + 28 = 0$

This simplifies to: $x^2 - 8x + 16 = 0$

$\Delta = b^2 - 4ac = (-8)^2 - 4(1)(16) = 0$

8 The line $2x - y + 3 = 0$ intersects the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ at two points, D and E.

a Find the coordinates of D and E.

Rearrange equation of line: $y = 2x + 3$

Sub into equation of circle

$x^2 + (2x + 3)^2 - 4x + 6(2x + 3) - 12 = 0$

This simplifies to $5x^2 + 20x + 15 = 0$

$\therefore (x + 1)(x + 3) = 0$

The points of intersection are $(-1, 1), (-3, -3)$

b Hence find the length of the line segment joining DE.

$DE = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{2^2 + 4^2} = 2\sqrt{5}$ units

9 For the circle $x^2 + y^2 = 4$ and line $y = mx + 4$, determine the exact values of the gradient, m, so that the line:

a is a tangent to the circle (intersects the circle once)

set up equation $x^2 + (mx + 4)^2 = 4$

$\Delta = 16m^2 - 48$

Let $\Delta = 0$

$m = \pm\sqrt{3}$

b intersects the circle at two points

$\Delta > 0$

$16m^2 - 48 > 0$

$m < -\sqrt{3}$ or $m > \sqrt{3}$

c does not intersect the circle

$-\sqrt{3} < m < \sqrt{3}$

10

a Try to find the coordinates of the points of intersection of $y = \frac{1}{x}$ and $y = -x + 1$.

What do you notice?

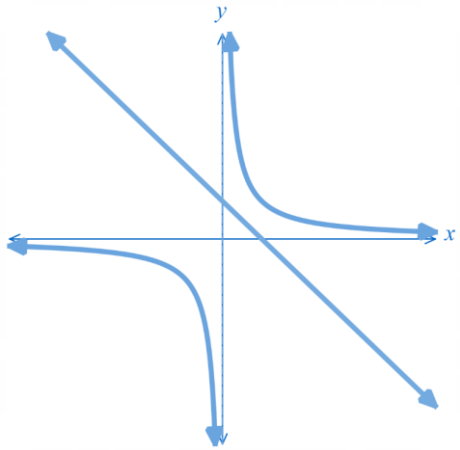
$-x + 1 = \frac{1}{x}$

$-x^2 + x = 1$

$x^2 - x + 1 = 0$

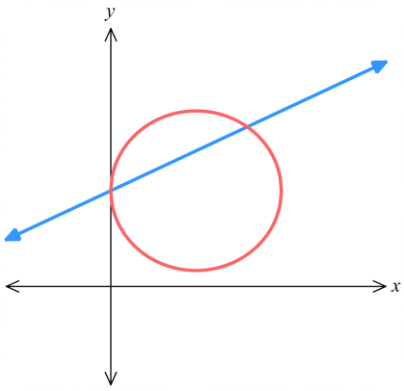
There are no points of intersection; $\Delta = (-1)^2 - 4(1)(1) = -3 < 0$.

b Sketch the two graphs on the same axis to visualise the scenario.



11 The line $2y - x = 12$ intersects the circle $x^2 + y^2 - 10x - 12y + 36 = 0$ at the points A and B.
A sketch is given, it would be useful to add to it as you go through parts a to d.

a Find the coordinates of A and B.



Using same method as previous question,

$(0,6)$ and $(8,10)$

b Find the equation of the perpendicular bisector of AB.

The perpendicular bisector of AB is the perpendicular line passing through the midpoint of AB.

Midpoint of AB: $\left(\frac{0+8}{2}, \frac{6+10}{2}\right) = (4,8)$

Gradient of AB: $m = \frac{10-6}{8-0} = \frac{1}{2}$

$\therefore$ gradient of perpendicular bisector is -2

Using $m = -2$ and $(4,8)$: $y - y_1 = m(x - x_1)$

$y - 8 = -2(x - 4)$

$y = -2x + 16$

c The perpendicular bisector of AB intersects the circle at the points P and Q.
Find the exact coordinates of P and Q.

Using same method as part **a**, $(5 + \sqrt{5}, 6 - 2\sqrt{5})$ and $(5 - \sqrt{5}, 6 + 2\sqrt{5})$

d Explain why APBQ is a kite and hence find its area.

The diagonals meet at right angles, and $PA = PB$ and $QA = QB$, so it satisfies the definition of a kite.

Using distance formula:

Length of AB is: $4\sqrt{5}$

Length of PQ is: 10

$A = \frac{xy}{2} = \frac{4\sqrt{5}(10)}{2} = 20\sqrt{5}$

12 Mr Ding used ChatGPT to generate an exam question:
'Find the points of intersection between $y = 2^x$ and $y = x^2 + 1$'
He checked on DESMOS and saw two 'nice' points of intersection at $(0,1)$ and $(1,2)$.
However, his students would not be able to find these points algebraically. Explain why.

It is not possible to solve $2^x = x^2 + 1$ using any standard algebraic techniques.

13 [Auxiliary circle of a hyperbola]
The equation $(x + y)^2 + (x - y)^2 = a^2$ is an equation representing a circle.

a By expanding then completing the square, find the radius of a circle in this form, in terms of a.

$r = \frac{a}{\sqrt{2}}$

b What kind of curve is $(x + y)^2 - (x - y)^2 = a^2$? (hint: expand and simplify)

Hyperbola with equation $y = \frac{a^2}{4x}$

c Use DESMOS to sketch $(x + y)^2 + (x - y)^2 = a^2$ and $(x + y)^2 - (x - y)^2 = a^2$.
Create a slider for a and experiment with values of a.
https://www.desmos.com/calculator/luwgidnzzy
What do you think the coordinates of the point of intersection are?

The points of intersection are $\left(\pm\frac{a}{2}, \pm\frac{a}{2}\right)$

d Verify this algebraically.

Start by expanding out each equation, after fully simplifying you should have $2x^2 + 2y^2 = a^2$ and $4xy = a^2$

Substitute $y = \frac{a^2}{4x}$ into the first equation to get $32x^4 - 16a^2x^2 + 2a^4 = 0$

Divide by the common factor and recognise as a perfect square: $(4x^2 - a^2)^2 = 0$

Solve to get $x = \pm\frac{a}{2}$

Substitute into a previous equation and solve for y: $y = \pm\frac{a}{2}$

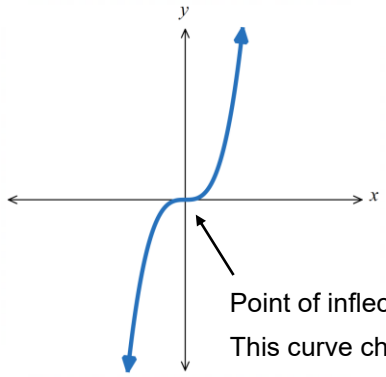
Cubic Curves and Transformations

ⓘ Cubic Curves

A cubic function has an equation where x has a highest power of 3.

The most basic form is $y = kx^3$

The basic cubic curve has a **point of inflection**, where the **concavity changes**.



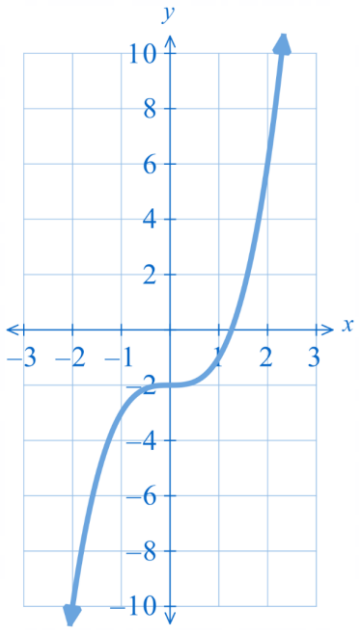
Point of inflection

This curve changes from concave down to concave up

ⓘ **Investigation** Graph cubic relationships using a table of values

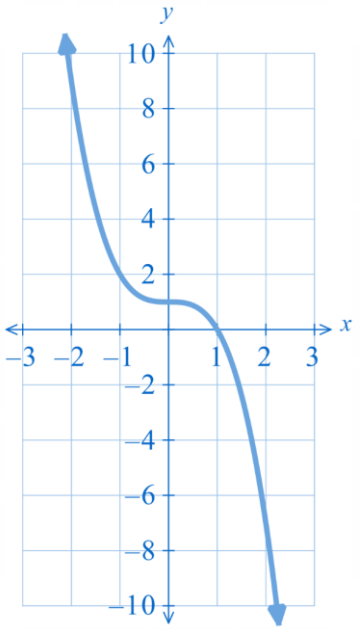
Draw the graph of $y = x^3 - 2$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							



Draw the graph of $y = -x^3 + 1$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							



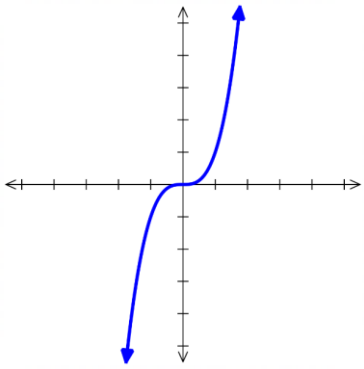
ⓘ Cubic Curve Transformations

For a cubic curve $y = k(x - b)^3 + c$

The parameters b, c, k affect the graph in predictable ways.

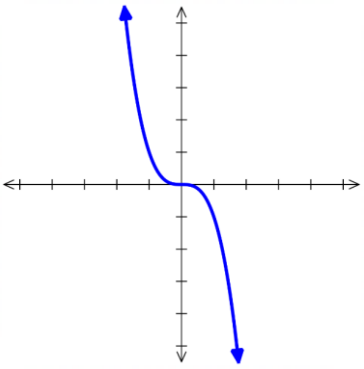
$k > 0$, the curve is increasing.

$y = 1x^3$



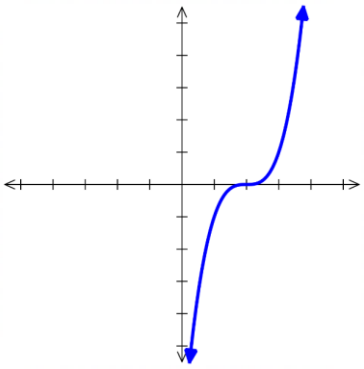
$k < 0$, the curve is decreasing

$y = -1x^3$



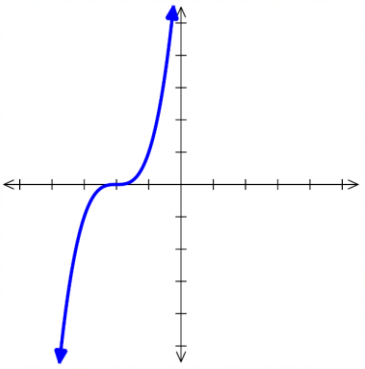
$b > 0$, the curve is shifted ……

$y = (x - 2)^3$



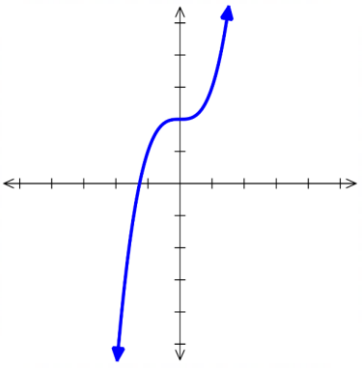
$b < 0$, the curve is shifted ……

$y = (x + 2)^3$



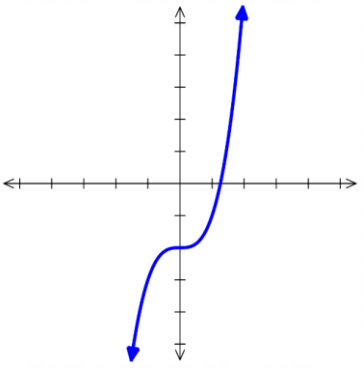
$c > 0$, the curve is shifted ……

$y = x^3 + 2$



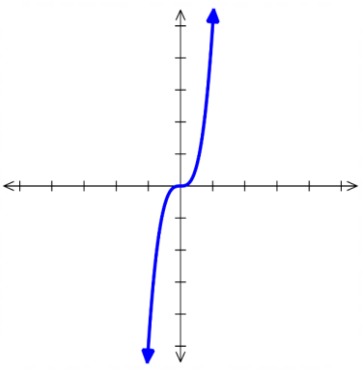
$c < 0$, the curve is shifted ……

$y = x^3 - 2$



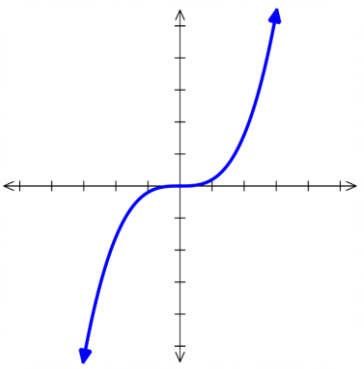
$|k| > 1$, the curve gets narrower.

$y = 5x^3$



$0 < |k| < 1$ the curve gets wider.

$y = \frac{1}{5}x^3$



Inflection Point Form of a Cubic Curve

The **inflection point form** of a cubic reveals its **inflection point** because it shows the graph's horizontal and vertical shifts.

$$y = a(x-h)^3 + k$$

Concavity — inflection point (h,k)

To sketch a cubic curve:

1. Determine the behaviour of curve by substituting $x \to \infty$.
2. Find y-intercept by letting $x = 0$.
3. Find x-intercepts by letting $y = 0$.
4. Identify the point of inflection (h,k).

Example Success Criteria

Sketch $y = x^3 - 8$

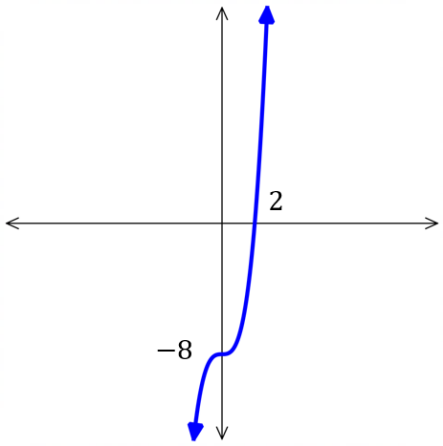
1. Behaviour — When $x \to \infty, y \to \infty$
2. y-int — Let $x = 0$

$$y = 0^3 - 8$$
$$= -8$$

3. x-int — Let $y = 0$

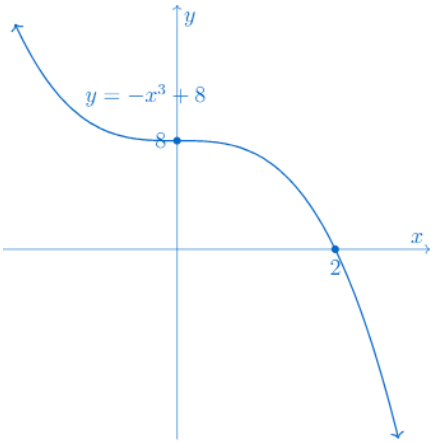
$$0 = x^3 - 8$$
$$8 = x^3$$
$$2 = x$$

4. Point of inflection at $(0,-8)$



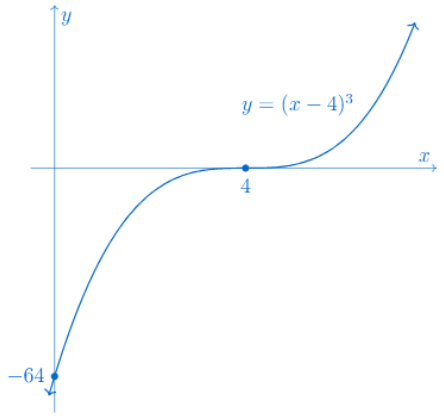
Guided Practice

Sketch $y = -x^3 + 8$

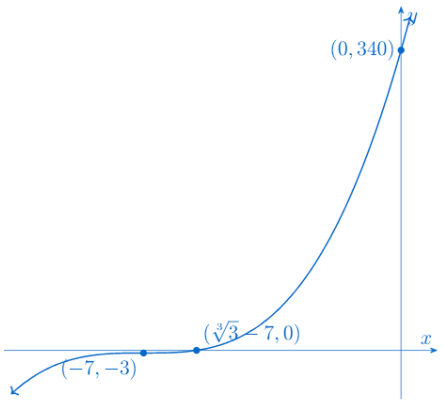


Guided Practice

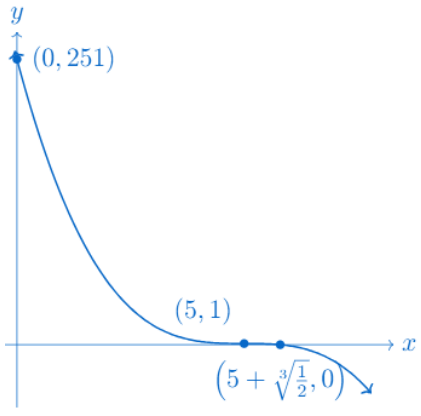
Sketch $y = (x-4)^3$



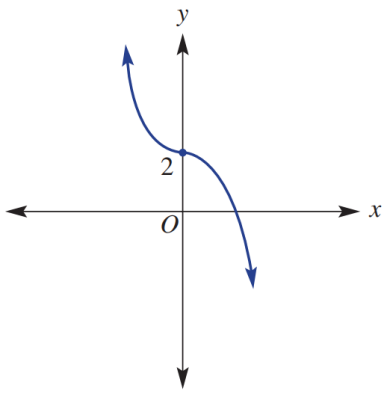
Sketch $y = (x+7)^3 - 3$



Sketch $y = -2(x-5)^3 + 1$



1 The graph of a cubic function is shown.



a Identify the point of inflection.

(0,2)

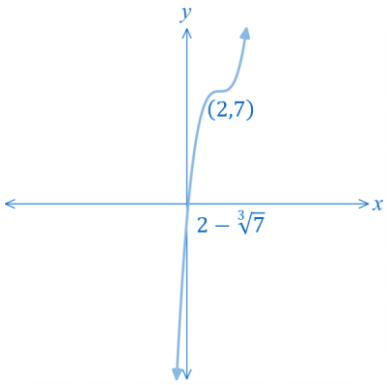
b Circle a possible equation for this cubic equation.

A $y = 2x^3$ **B** $y = (x+2)^3$
C $y = -2x^3$ **D** $y = -x^3 + 2$
E $y = x^3 + 2$

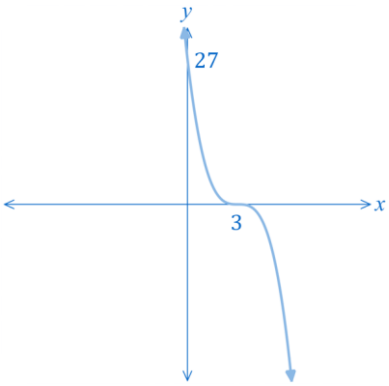
D

2 Draw a neat sketch of these equations.
Clearly label the y-intercept, x-intercept.

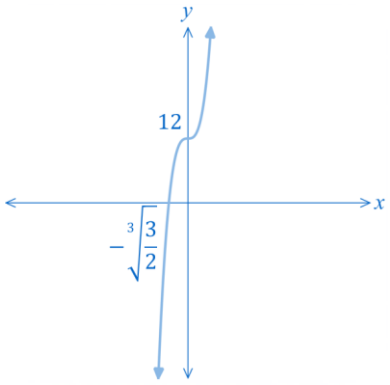
a $y = (x-2)^3 + 7$



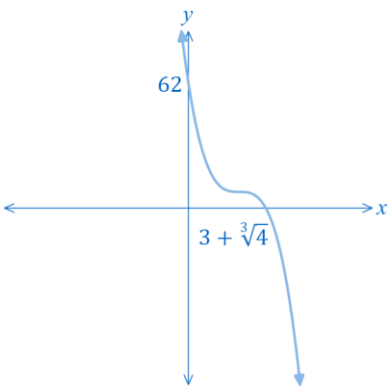
b $y = -(x-3)^3$



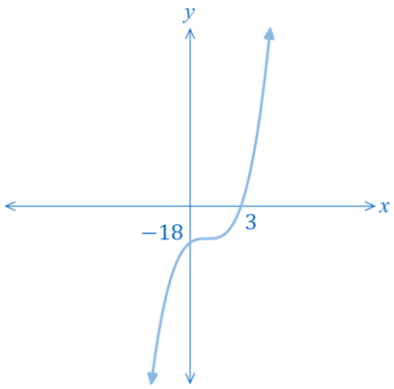
c $y = 8x^3 + 12$



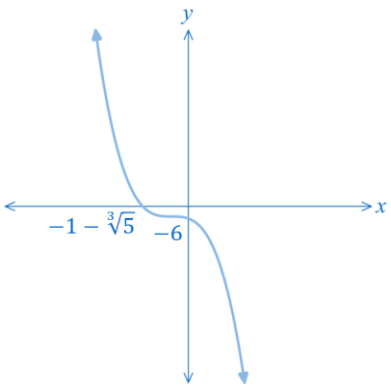
d $y = -2(x-3)^3 + 8$



e $y = 2(x-1)^3 - 16$

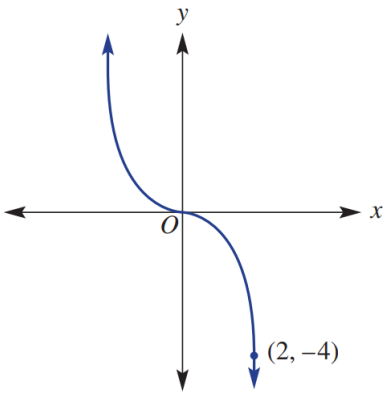


f $y = -(x+1)^3 - 5$



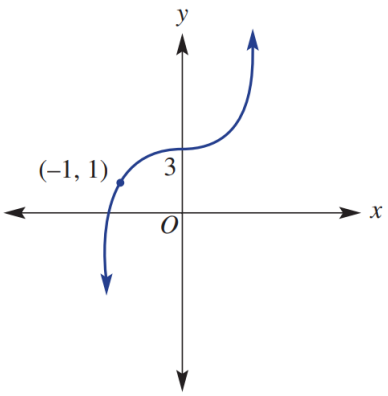
3 The cubic curves are in the form $y = kx^3 + c$. Find its equation.

a



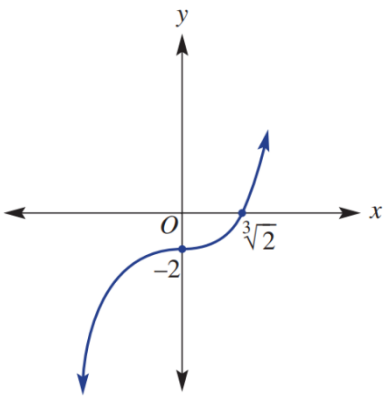
$y = -\frac{x^3}{2}$

b



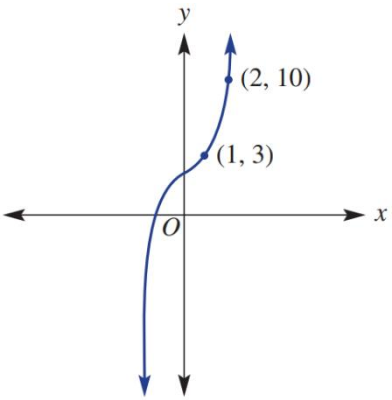
$y = 2x^3 + 3$

c



$y = x^3 - 2$

d



$y = x^3 + 2$

Power Curve Transformations

ⓘ Power Curve Transformations

For a power curve $y = k(x - b)^n + c$

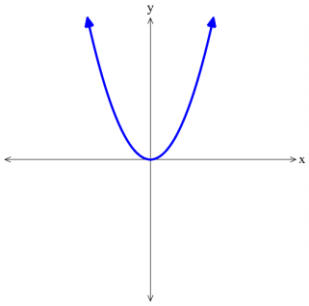
The parameters b, c, k, n affect the graph in predictable ways.

n is even, curve has a turning point.

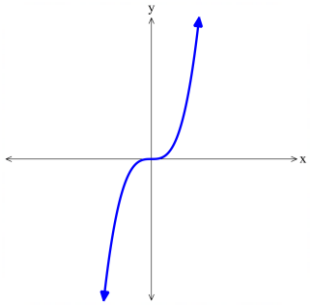
n is odd, curve has inflection point.

As n increases, the curve gets …… but the turning/inflection point gets ……

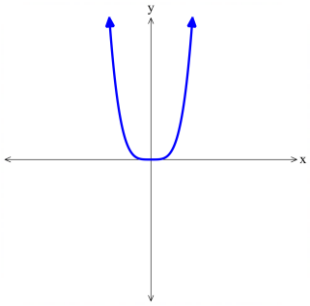
$y = x^2$



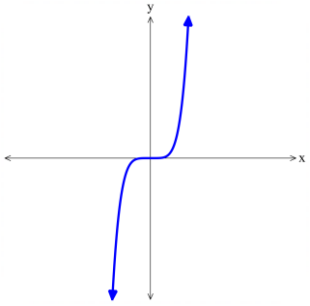
$y = x^3$



$y = x^4$

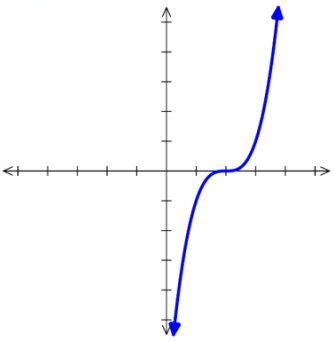


$y = x^5$



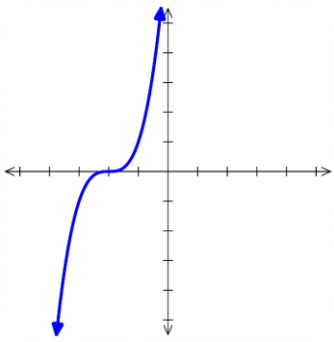
$b > 0$, the curve is shifted ……

$y = (x - 2)^3$



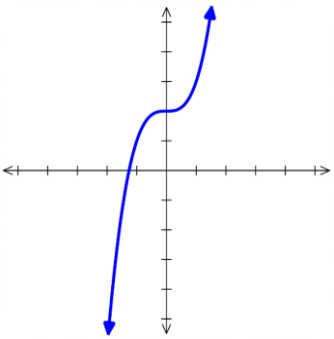
$b < 0$, the curve is shifted ……

$y = (x + 2)^3$



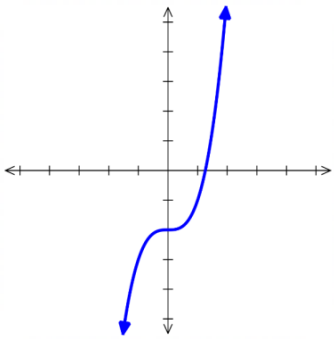
$c > 0$, the curve is shifted ……

$y = x^3 + 2$



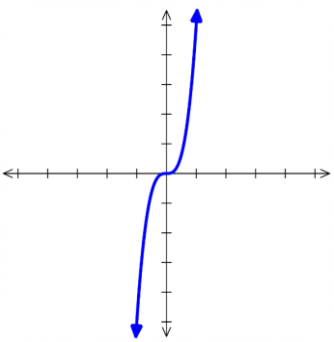
$c < 0$, the curve is shifted ……

$y = x^3 - 2$



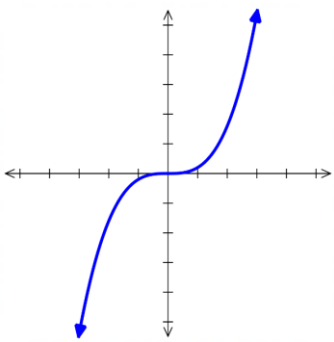
$|k| > 1$, the curve gets narrower.

$y = 5x^3$



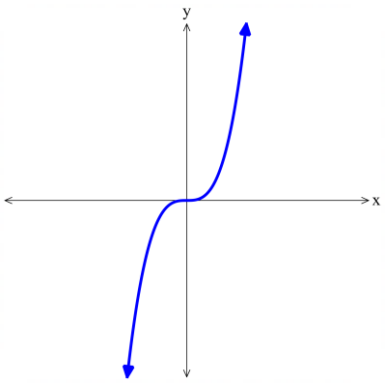
$0 < |k| < 1$ the curve gets wider.

$y = \frac{1}{5}x^3$



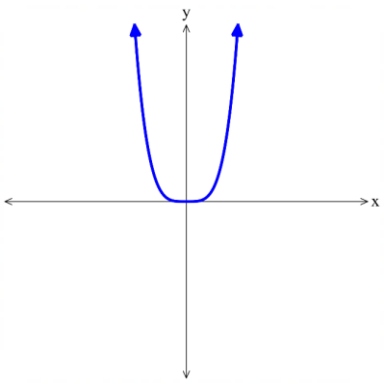
1 The graph of $y = x^n$ is given. State whether the power is even or odd.

a



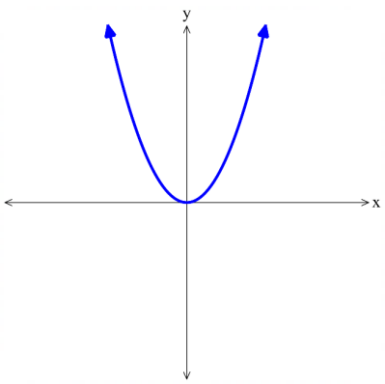
odd

b



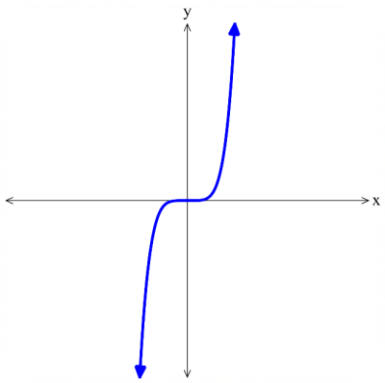
even

c



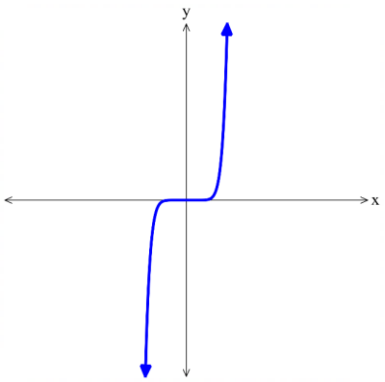
even

d



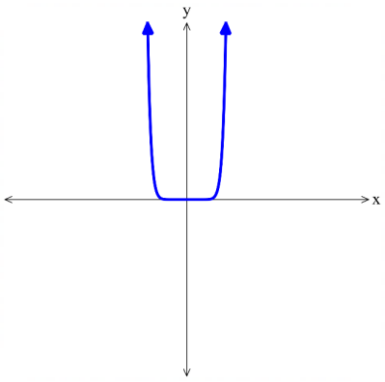
odd

e



odd

f

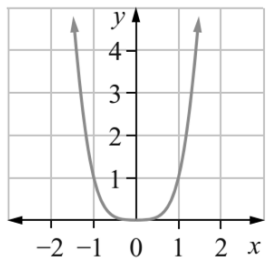


even

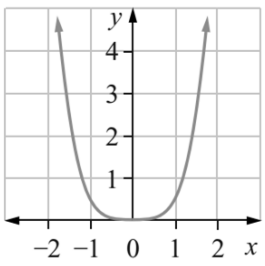
2 Match each graph with its equation.

a $y = x^4$ | **b** $y = 5x^4$ | **c** $y = \frac{1}{2}x^4$ | **d** $y = \frac{x^4}{6}$

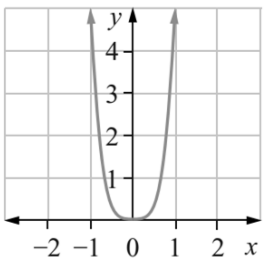
A



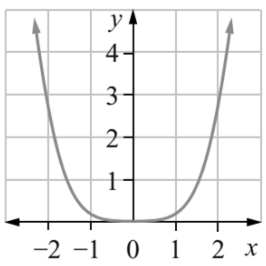
B



C



D

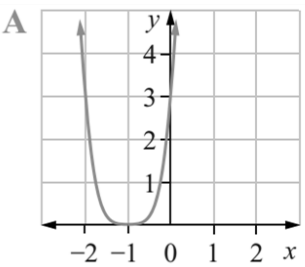


a – A, b – C, c – B, d - D

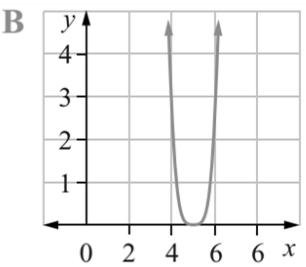
3 Match each graph with its equation.

a $y = 3(x-1)^4$ | **b** $y = 3(x+1)^4$ | **c** $y = 3(x+3)^4$ | **d** $y = 3(x-5)^4$

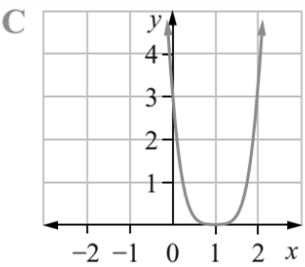
A



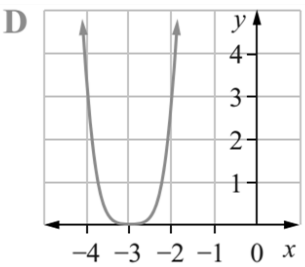
B



C



D

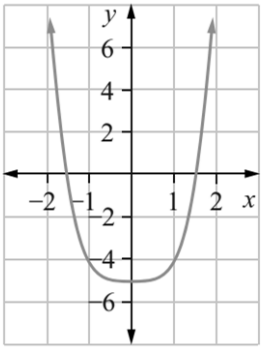


a – C, b – A, c – D, d – B

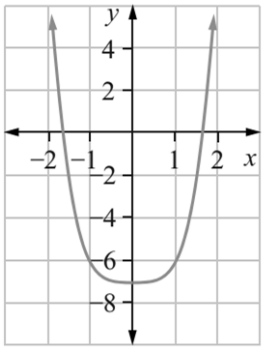
4 Match each graph with its equation.

a $y = x^4 - 5$ | **b** $y = x^4 + 1$ | **c** $y = x^4 - 7$ | **d** $y = x^4 + 5$

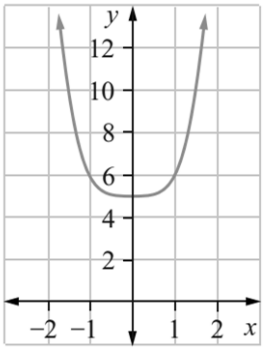
A



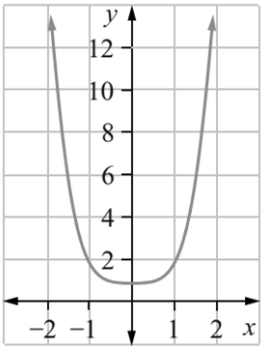
B



C



D



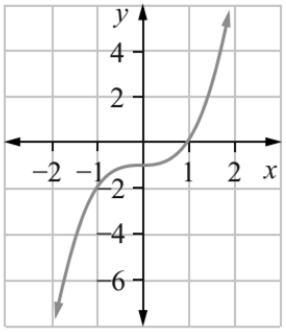
a – A, b – D, c – B, d – C

5 Match each graph with its equation.

$y = 4x^2$ $y = -4x^2$ $y = x^3 - 1$

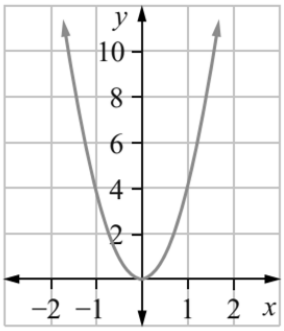
$y = -x^3 + 1$ $y = (x + 2)^6$ $y = -(x + 2)^6$

a



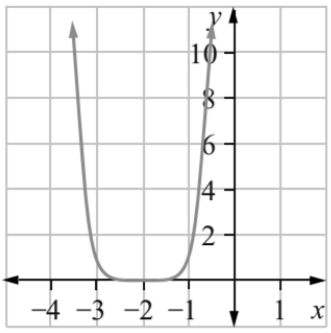
$y = x^3 - 1$

b



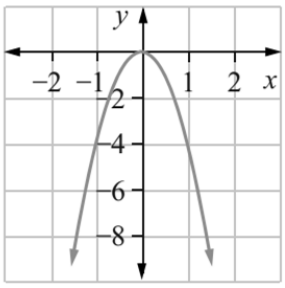
$y = 4x^2$

c



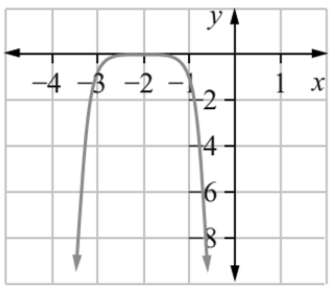
$y = (x + 2)^6$

d



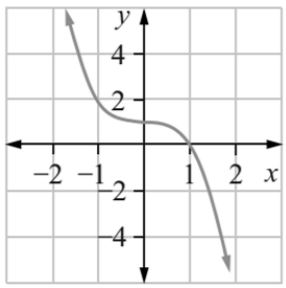
$-4x^2$

e



$y = -(x + 2)^6$

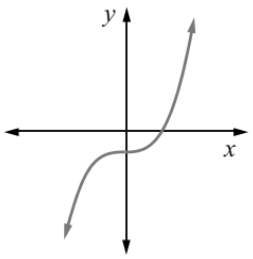
f

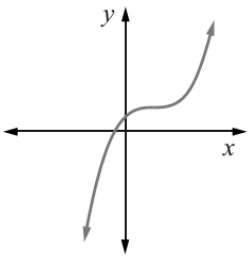


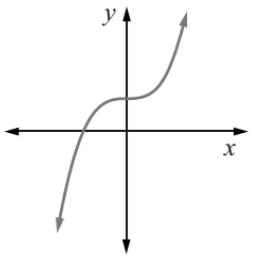
$y = -x^3 + 1$

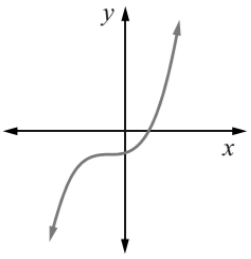
6 Which of the following could be the graph of each function?

a $y = (x+3)^3 - 2$

A 

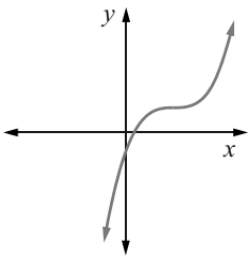
B 

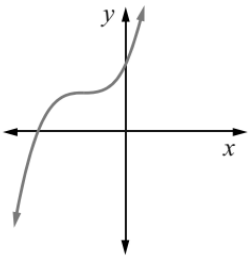
C 

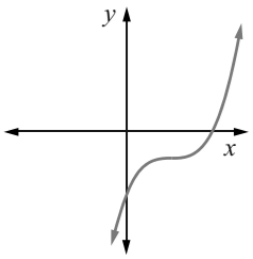
D 

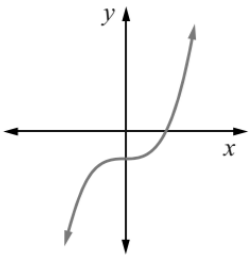
D

b $y = (x+3)^3 + 4$

A 

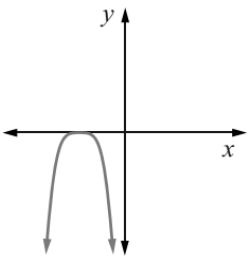
B 

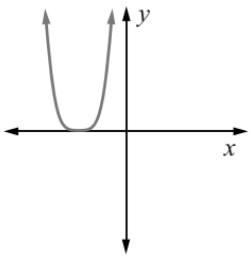
C 

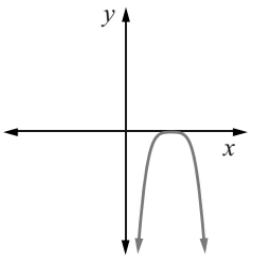
D 

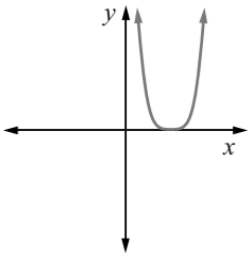
B

c $y = -(x-3)^4$

A 

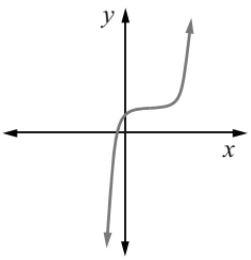
B 

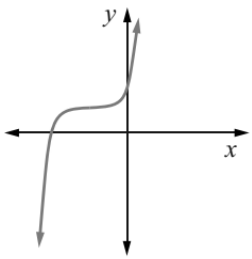
C 

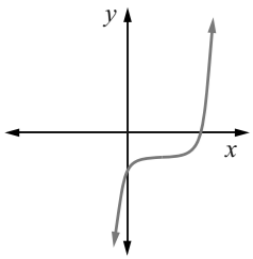
D 

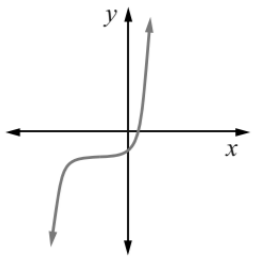
C

d $y = (x+3)^5 - 2$

A 

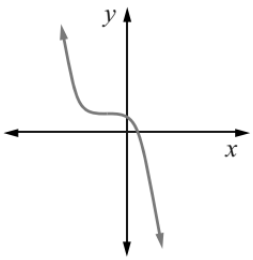
B 

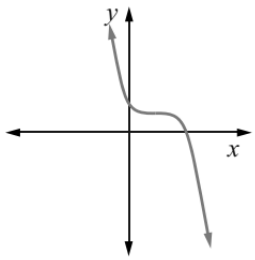
C 

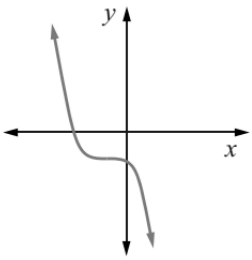
D 

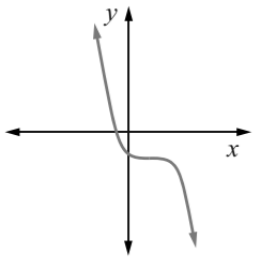
D

e $y = -(x+2)^3 + 1$

A 

B 

C 

D 

A

Sketching Cubics in Factored Form

ⓘ Leading Term of a Cubic

For a cubic equation $y = ax^3 + bx^2 + cx + d$, $\boldsymbol{ax^3}$ is called the **leading term.**

It has the biggest impact on the end behaviour of the curve.

The **constant term**, $\boldsymbol{d}$, will tell us the y-intercept of the curve.

ⓘ **Identify** the leading term of a cubic

To find the leading term, multiply the highest power terms from each factor.

To find the constant term, multiply the constant terms of each factor.

$$y = (2x - 3)(x - 4)(x + 5)$$

Leading term: $2x^3$

Constant term: 60

$$y = (3 - 2x)(x - 4)(x + 5)$$

Leading term: $-2x^3$

Constant term: -60

Find the leading term and constant term of these cubics.

a $y = (x + 2)(x - 1)(x - 3)$

b $y = -(x - 2)(x - 1)(x - 3)$

c $y = x(x + 5)(2x + 3)$

d $y = -3x(x - 5)(2x + 15)$

e $y = (3 - x)(2 + x)(x + 5)$

f $y = (2x - 3)(3x + 1)(7 - x)$

ⓘ Sketching a Cubic Curve in Factored Form

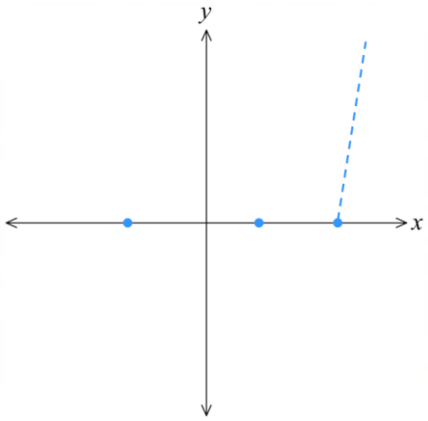
x-intercepts

$$y = k(x-\alpha)(x-\beta)(x-\gamma)$$

1. Find the leading term and determine the end behaviour as $x \to \infty$.
2. Find x intercepts.
3. Find y intercept.

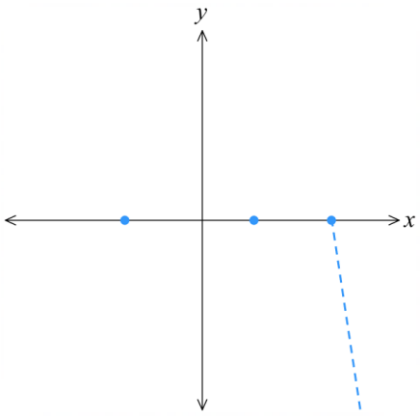
Leading term positive

As $x \to \infty$, $y \to \infty$



Leading term negative

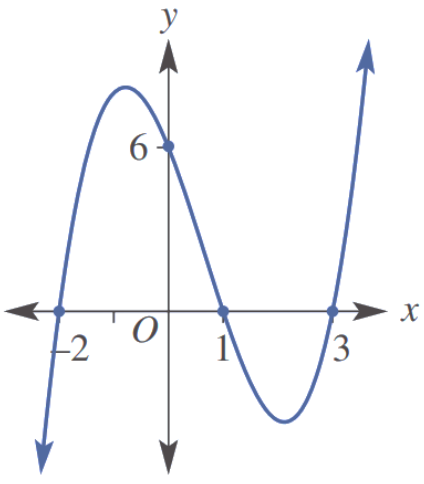
As $x \to \infty$, $y \to -\infty$



Example Sketch cubic curves in factored form

Sketch $y = (x+2)(x-1)(x-3)$

1. The leading term is x^3,
   so when $x \to \infty$, $y \to \infty$
2. Let $y = 0$
   $0 = (x+2)(x-1)(x-3)$
   $x = -2, 1, 3$
3. Let $x = 0$
   $y = (0+2)(0-1)(0-3)$
   $= (2)(-1)(-3)$
   $= 6$



Guided Practice

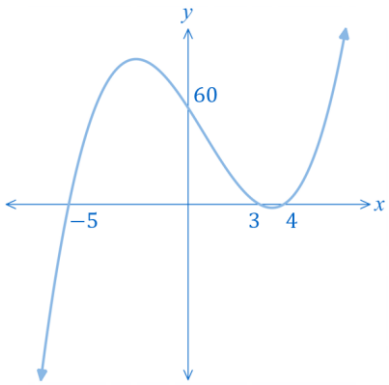
Sketch these cubic curves.

a $y = (x - 4)(x + 5)(x - 3)$

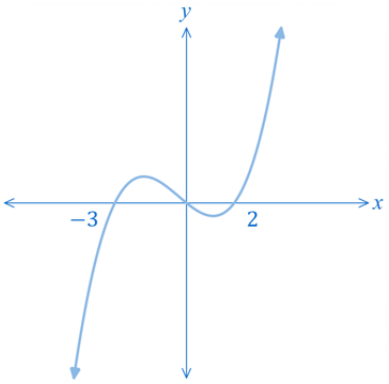
1. leading term & behaviour

2. x-intercepts

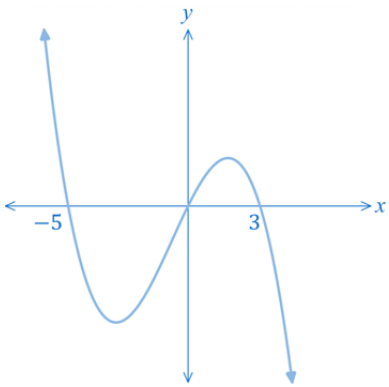
3. y-intercept



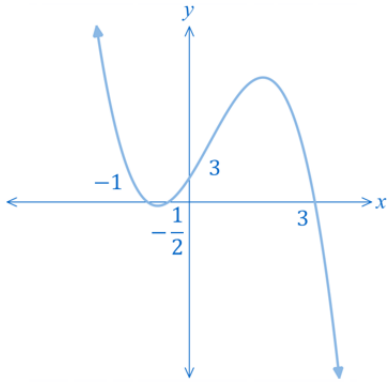
b $y = x(x - 2)(x + 3)$



c $y = (x - 4)(x + 5)(x - 3)$

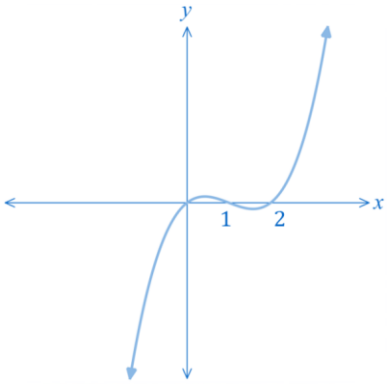


d $y = (3 - x)(x + 1)(2x + 1)$

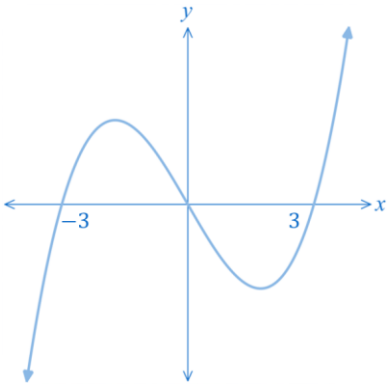


1 Sketch these cubic curves. Show the x and y-intercepts.

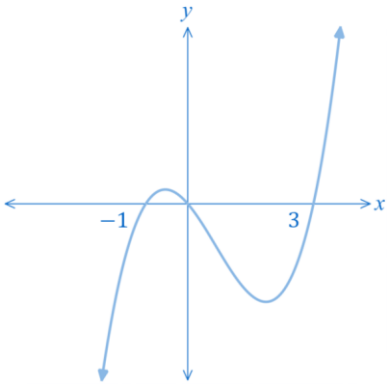
a $y = x(x-1)(x-2)$



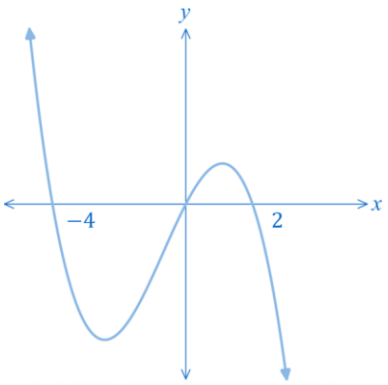
b $y = x(x-3)(x+3)$



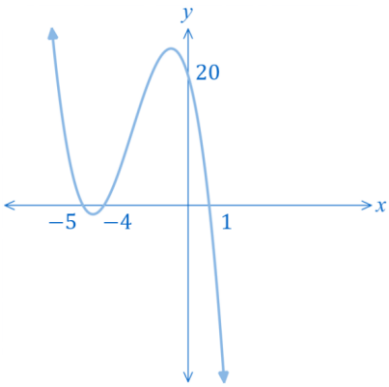
c $y = 2x(x-3)(x+1)$



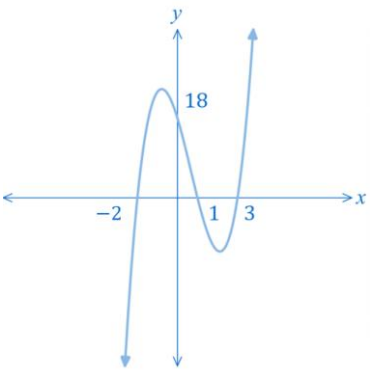
d $y = -x(x-2)(x+4)$



e $y = -(x+4)(x-1)(x+5)$

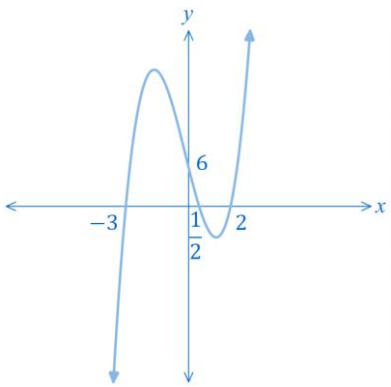


f $y = 3(x+2)(x-1)(x-3)$

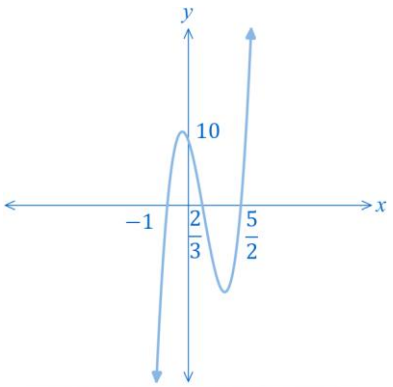


2 Sketch these cubic curves. Show the x and y-intercepts.

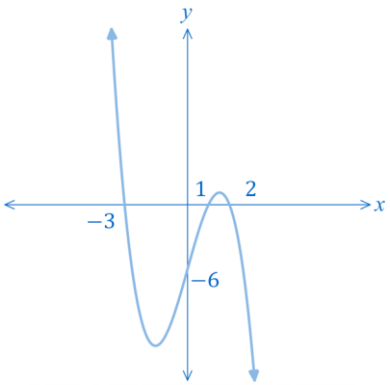
a $y = (2x-1)(x+3)(x-2)$



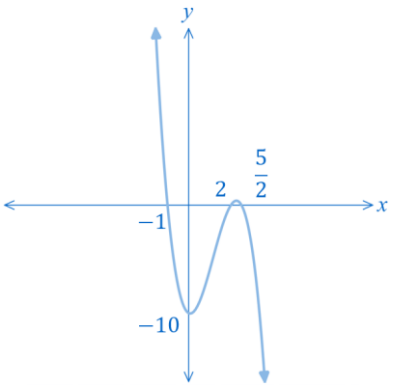
b $y = (3x-2)(x+1)(2x-5)$



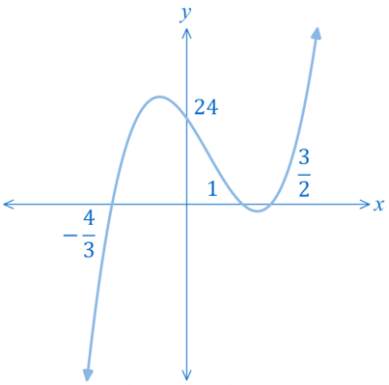
c $y = (1-x)(x+3)(x-2)$



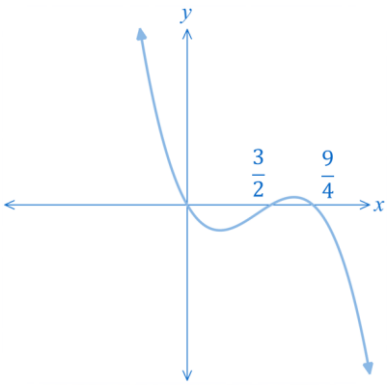
d $y = (2-x)(x+1)(2x-5)$



e $y = -2(3x+4)(2x-3)(1-x)$

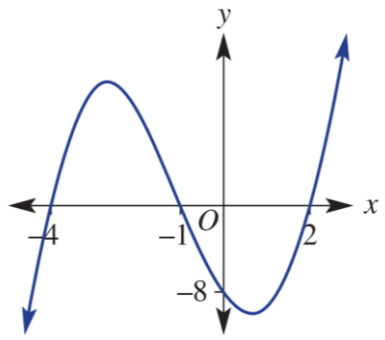


f $y = -x(9-4x)(3-2x)$



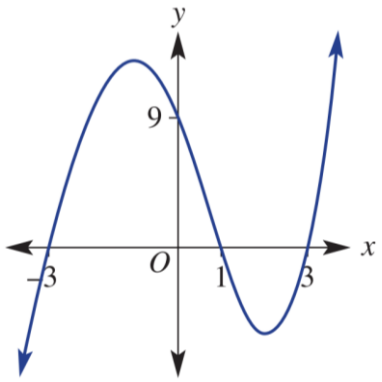
3 The following graphs are cubic curves. Find the equation using $y = k(x - \alpha)(x - \beta)(x - \gamma)$.

a



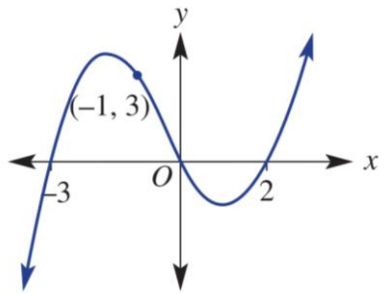
$y = (x - 2)(x + 1)(x + 4)$

b



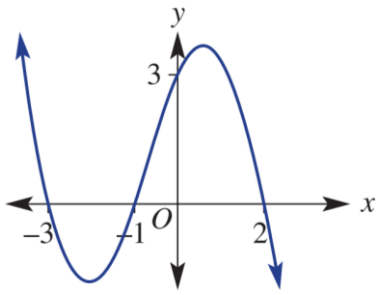
$y = (x + 3)(x - 1)(x - 3)$

c



$y = \frac{1}{2}x(x - 2)(x + 3)$

d



$y = -\frac{1}{2}(x + 3)(x + 1)(x - 2)$

4 A cubic function is in the form $y = k(x - \alpha)(x - \beta)(x - \gamma)$. Find its equation if:

a it has x-intercepts 2, 3 and −5 and passes through the point (−2, –120).

$y = -2(x - 2)(x - 3)(x + 5)$

b it has x-intercepts −1, 4 and 6 and passes through the point (3, 96).

$y = 8(x + 1)(x - 4)(x - 6)$

c it has x-intercepts 1 and 3, y-intercept −27 and k = −3.

$y = -3(x - 1)(x - 3)(x + 3)$